UL TEST REPORT

Prepared For :	Oudian(beijing)Electric Appliance Co.,LTD Room 1805,floor 18,Building 9,Yard 25,Daxing Jiuqiao Road,Economic and Technological Development Zone,Beijing ,P.R.China
Product Name:	HOT AIR STYLER
Model :	OD-M6808
Prepared By :	Shenzhen HTT Technology Co., Ltd. 1F, B Building, Huafeng International Robotics Industrial Park, Gushu, Xixiang Street, Bao'an District, Shenzhen
Test Date:	Apr. 29, 2025 ~ May. 08, 2025
Date of Report :	May. 08, 2025
Report No.:	HTT202505054LR

UL TEST REPORT
UL 859
Household Electric Personal Grooming Appliances

Testing Laboratory	
Name.. :	Shenzhen HTT Technology Co., Ltd.
Address.. :	1F, B Building, Huafeng International Robotics Industrial Park, Gushu, Xixiang Street, Bao'an District, Shenzhen
Testing location...................................:	Shenzhen HTT Technology Co., Ltd.
Applicant...:	Oudian(beijing)Electric Appliance Co.,LTD
Address.. :	Room 1805,floor 18,Building 9,Yard 25,Daxing Jiuqiao Road,Economic and Technological Development Zone,Beijing ,P.R.China
Test specification:	UL test report
Standard...:	UL 859 (11th ed., Rev. June.20,2012)
Test procedure.................................... :	Comply with the Standard UL 859 (11th ed., Rev. June.20,2012)
Non-standard test method…………..:	N/A
Product ……………………………..…….:	HOT AIR STYLER
Trade Mark..:	OUDIM
Model and/or type reference..............:	OD-M6808
Manufacturer .... :	OUDIM LLC
Address ………………………………:	18012 COWAN #231 IRVINE, CA 92614
Rating(s)..:	Input: 125V~, 60Hz Power: 1200W
Test item particulars	
Classification of installation and use:	Class II
Supply Connection.............................. :	AC plug of power wire connection
Possible test case verdicts	
- test case does not apply to the test object:	N/(N/A)
- test object does meet the requirement:	P(Pass)
- test object does not meet the requirement:	F(Fail)

Name and address of the testing laboratory: Shenzhen HTT Technology Co., Ltd.
1F, B Building, Huafeng International Robotics Industrial Park, Gushu, Xixiang Street, Bao'an District, Shenzhen

Test by : Darek Wang
Signature

May. 08, 2025
Date

Review by : Andy Shen
Signature

May. 08, 2025
Date

Approved by : Kevin Yang
Signature

SHENZHEN HTT TECHNOLOGY CO., LTD
HTT

May. 08, 2025
Date

KevinYang/ Manager
Name and Title

General remarks

This test report shall not be reproduced except in full without the written approval of the testing laboratory.

The test results presented in this report relate only to the item tested.

"(see remark #)" refers to a remark appended to the report.

"(see appended table)" refers to a table appended to the report.

General product information:

Copy of marking plate(s):

Summary of testing:

The sample(s) tested comply with the requirement of UL 859 (11th ed., Rev. June.20,2012).

Model List:	
Test Model	OD-M6808
1. All tests are carried out on OD-M6808.	

UL 859			
Clause	Requirement + Test	Result - Remark	Verdict

CONSTRUCTION			
6	**General**		**P**
6.1	In the following text, a requirement that applies only to a specific type or types of appliances, such as a hand-supported hair dryer and a curling iron, is so identified by specific reference in that requirement to the type or types involved. Absence of such specific reference or use of the term ″appliance″indicates that the requirement applies to all appliances covered by this standard.	Complied.	P
6.2	An appliance that is a combination of two or more types (for example, an appliance having a hand-supported part and a counter-supported part), or an appliance that fits the definition of two or more types (for example, an appliance that can be used while supported by hand or while supported by a counter top), is to be investigated in accordance with the applicable requirements for the types of appliances involved. If two requirements that address the same condition differ, the appliance is to be investigated to the more severe requirement.		N
6.3	A heated air curling iron or brush, as defined in 5.25, shall comply with the requirements applicable to hand-supported hair dryers and curling irons.		P
6.4	A container for liquid intended for use with the appliance, and supplied as part of the appliance, shall comply with applicable construction requirements.		N
6.5	A curling iron that is likely to be laid on combustible material shall be provided with a stand made of material resistant to combustion upon which it may be placed when not in use.		N
6.6	A curling iron that attains a temperature higher than 100°C (212°F) when operated continuously shall be provided with an integral stand. A stand provided for other types of appliances may be a separate device or integral with the appliance.		N
6.7	With respect to 6.6, an integral stand provided for a curling iron shall be of such design or shape that any surface of the curling iron exceeding 150°C (302°F) will not contact the supporting surface when the curling iron is supported in its intended manner by the stand.		N
6.8	A polymeric material used as an integral stand in compliance with the requirements in 6.7 shall be rated for the temperature it is subjected to during use.		N

7	**Hair Dryer Immersion Protection**		P
7.1	A hand-supported hair-drying appliance (such as a hair dryer, blower-styler, styler-dryer, heated air comb, heated-air hair curler, curling iron-hair dryer combination, a wall-hung hair dryer or the hand unit of a wall-mounted hair dryer, or a similar appliance) shall be constructed to Reduce the risk of electric shock when the appliance is energized, with its power switch in either the″on″or″off″position, and immersed in water having an electrically conductive path to ground.		P

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Clause	Requirement + Test	Result - Remark	Verdict

7.2	Compliance with 7.1 may be accomplished with the use of an:		N
	a) Integral ground-fault circuit-interrupter (GFCI) or		N
	b) Integral protective device of another type that de-energizes all current-carrying parts (hereafter referred to as a protective device) when the hand-supported hair-drying appliance is immersed in water having an electrically conductive path to ground.		N
7.3	If a hand-supported hair-drying appliance is provided with a GFCI, the GFCI shall comply with the requirements for Class A cord-connected GFCIs in the Standard for Ground-Fault Circuit-Interrupters, UL 943.		N
7.4	If a hand-supported hair-drying appliance is provided with a protective device other than a GFCI, the protective device shall be investigated and determined to be acceptable for the application. Investigation of the protective device shall include, but need not be limited to		P
7.5	A GFCI or other protective device shall be integral with the attachment plug of the hand-supported hair-drying appliance power supply cord.		P
7.6	A user-resettable protective device shall incorporate a supervisory circuit as described in the Standard for Ground-Fault Circuit-Interrupters, UL 943, for GFCIs.		N
7.7	A switch included for testing a user resettable protective device shall be permanently marked″Test″ and″Reset ″on or adjacent to the switch actuators.		P
7.8	After a protective device de-energizes current-carrying parts, it shall not automatically reset.		P
7.9	A protective device that is integral with the attachment plug of a hand-supported hair-drying appliance may be provided with a single outlet convenience receptacle		N
7.10	With regard to 7.9(f), each output circuit shall be considered if one is not representative of the other. For example, the short circuit test shall be conducted with each output short-circuited one at a time. The dielectric voltage-withstand test between line-connected circuits and load circuits shall include both load circuits. The temperature test shall be conducted with:		P
	a) The hair dryer load circuit and the convenience receptacle each loaded to rated value and		N
	b) The convenience receptacle loaded to 15 amperes with no load connected to the hair dryer load circuit.		P

8	**Frame and Enclosure**		**P**
8.1	General		P
8.1.1	The frame and enclosure of an appliance shall be sufficiently strong and rigid to resist the abuses likely to be encountered during service. The degree of resistance inherent in the appliance shall preclude total or partial collapse with the attendant reduction of spacings, loosening or displacement of parts, and other conditions which alone or in combination constitute an increase in the risk of fire, electric shock, or injury to persons.	No any risks.	P

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Clause	Requirement + Test	Result - Remark	Verdict
8.1.2	Among the factors taken into consideration in evaluating an enclosure for acceptability are its:	Complied.	P
	a) Physical strength,		P
	b) Resistance to impact,		P
	b) Resistance to impact,		P
	d) Combustibility,		P
	e) Resistance to corrosion, and		N
	f) Resistance to distortion at temperatures to which the enclosure may be subjected under conditions of normal or abnormal use.		P
	For a nonmetallic enclosure, all these factors are to be considered with respect to thermal aging.		P
8.2	Polymeric enclosures and parts	Complied.	P
8.2.1	A polymeric enclosure shall comply with the Standard for Polymeric Materials – Use in Electrical Equipment Evaluations, UL 746C.		P
8.3	Metal enclosures	No metal enclosure used.	N
8.3.1	The minimum thickness of a metal enclosure shall be as indicated in Table 8.1.		N
8.4	Corrosion resistance		N
8.4.1	Iron and steel parts shall be made corrosion resistant by painting, galvanizing, plating, or other equivalent means if the malfunction of such unprotected parts would result in a risk of fire, electric shock, or injury to persons.		N
8.4.2	A container for liquid shall be made resistant to the possible corrosive effect of the liquid intended to be used in the container.		N
8.5	Accessibility of live parts		P
8.5.1	An electrical part of an appliance shall be located or enclosed so that unintentional contact with any uninsulated live part and internal wiring will be prevented.	No any risks.	P
8.5.2	A part of the outer enclosure that is capable of being opened or removed by the user without using a tool (to attach an accessory, to make an operating adjustment, to replace a fuse, or for other reasons) is to be opened or removed when determining compliance with 8.5.1.	No any risks.	P
8.5.3	The enclosure of an appliance shall have no opening that permits a probe, as illustrated in Figure 8.1, to touch any part that involves a risk of electric shock.	No any risks.	P
8.5.4	With regard to 8.5.3, the probe is to be articulated into any configuration and rotated or angled to any position before, during, or after insertion into the opening. The penetration shall be to any depth allowed by the opening size, including minimal depth combined with maximum articulation. The probe shall be applied with the minimum force required to determine accessibility and not as an instrument to evaluate the strength of a material.		P
8.5.5	An opening that will permit entrance of a 1-inch (25.4-mm) diameter rod is permitted when it complies with the conditions shown in Figure 8.2.		N
8.5.6	A live part of a limited-energy circuit in 5.30 requires the		P

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Clause	Requirement + Test	Result - Remark	Verdict
	same degree of protection against unintentional contact as a live part of a line voltage circuit.		
8.5.7	Insulated brush caps do not require additional enclosure.		P
8.5.8	An area of an enclosure that is provided with a group of openings or with a guarded opening (such as a grille, louver, or screen) is to be subjected to the strength of enclosure test described in 35.1.		N
8.5.9	The enclosure of a remotely or automatically controlled appliance shall reduce the risk of molten metal, burning insulation, or flaming particles, from falling on combustible materials, including the surface upon which the appliance is supported.		N
8.5.10	The requirement in 8.5.9 will necessitate the use of a barrier of material that is resistant to combustion		N
8.5.11	The requirement in 8.5.9 will also necessitate that a switch, relay, solenoid, or the similar part be individually and completely enclosed unless there is no opening in the bottom of the appliance enclosure, or it can be shown that malfunction of the component would not result in a risk of fire.		N
8.5.12	The barrier specified in 8.5.10 shall be horizontal, shall be located as indicated in Figure 8.3, and shall have an area no less than that described in Figure 8.3. An opening such as for drainage or ventilation, is permitted in the barrier if such an opening would not permit molten metal, burning insulation, or flaming particles to fall on combustible material.		N
8.6	Doors and covers		N
8.6.1	The door or cover of an enclosure shall be provided with means for holding it in the closed position.		N
8.6.2	The door or cover of an enclosure shall be hinged (or similarly attached) if it gives access to any overload protective device, the functioning of which requires	No such protective device used.	N
	renewal, or if it is necessary to open the cover in connection with the operation of the protective device. Such a door or cover shall be provided with a latch or similar device and shall be tight-fitting or shall overlap the surface of the enclosure around the opening.		N

9	**Reduction of Risk of Injury to Persons**		**P**
9.1	General		P
9.1.1	Materials that are relied upon to reduce the risk of injury to persons shall have such properties as to meet the demand of intended loading conditions.		P
9.1.2	Asbestos shall not be used.		P
9.1.3	A moving part that can result in a risk of injury to persons shall be enclosed or provided with other means to reduce unintentional contact	No moving part used.	N
9.1.4	With respect to the requirement specified in 9.1.3, the construction and intended use of the appliance are to be considered in investigating a guard or enclosure.		N
9.1.5	An appliance, or any item furnished with an appliance, shall have no sharp edge, burr, point, or spike inside or outside the appliance that results in injury to persons during	No any risks.	P

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	intended use and maintenance.		
9.1.6	On an appliance adjustable for height, means shall be provided for holding the upper parts securely in position. Means shall also be provided to prevent the upper part from descending rapidly if the securing means loosens or fails to operate as intended.	No such parts used.	P
9.1.7	A hand-supported hair dryer shall have each air intake opening provided with a screen or equivalent means so that there are no openings larger than 0.004 square inch (0.03 cm2)		N
9.2	Appliances with reservoirs		N
9.2.1	An appliance in which liquid reaches a temperature greater than 46°C (114.8°F) shall comply with the requirements specified in 9.2.2 – 9.2.4, 36.1 – 36.3, and 37.2.		N
9.2.2	The construction of the appliance shall reduce the risk of injury to persons under conditions of intended use. Openings through which liquid can be emitted shall not be provided unless such openings are needed to perform an operating function.		N
9.2.3	An appliance with a vessel or container with a capacity of more than 32 fluid ounces (946 mL) shall be provided with a fully inserting or a lock-on lid.		N
9.2.4	If any part of an appliance requires assembly (for example, engagement of a twist-lock part), then improper assembly that results in a risk of injury to persons shall be clearly visible to the user.		N
9.3	Wax depilatory appliances	No Wax depilatory appliances.	N
9.3.1	The maximum temperature of the wax, measured as described in 44.2.1 – 44.2.3, shall not exceed 75°C (167°F).		N
9.3.2	The maximum temperature rise of surfaces that may be contacted by the user shall be as specified in Table 44.2.		N
9.3.3	When there are multiple heat settings (for example, a setting for maintaining molten wax at the intended temperature for application to all of the skin and a higher heat setting for quick melting of solid wax)		N
9.3.4	With reference to 9.3.3(a), a nonremovable cover is one which requires special tools (tools not available to other than service personnel) for removal. A self-closing cover is a cover that returns to its fully closed position without any action on the part of the user other than releasing it from any opened position while the appliance is supported by a flat, horizontal surface.		N
9.3.5	In accordance with 46.9.5 and 76.8, if the malfunction of a temperature-regulating control increases the application temperature of the wax above 75°C (167°F), visible means, such as an indicator light, shall be provided to inform the user of an overheat condition.		N

10	**Mechanical Assembly**		**P**
10.1	An appliance that involves a motor or other vibrating part shall be assembled such that the appliance will not be affected adversely by the vibration. Brush caps shall be tightly threaded or otherwise constructed to prevent loosening.		N

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Clause	Requirement + Test	Result - Remark	Verdict
10.2	A switch (other than a through-cord switch), lampholder, receptacle, motor-attachment plug, or similar component shall be mounted securely and shall be prevented from turning.		P
10.3	Friction alone shall not be relied on for turn-prevention as required in 10.2. A lock-washer, applied as intended, is a reliable means of turn-prevention of a device with a single-hole mounting means		P
10.4	A positive means shall be provided to prevent parts of an appliance from turning with respect to each other if such turning would result in reduction of spacings, twisting of wires, and the like		N
10.5	A fastener that secures the insulating tip of a curling iron, a heated brush, or a similar appliance shall be constructed, fastened, or located so as to prevent the fastener from becoming loosened if such loosening can result in a risk of fire or electric shock.	No risk of fire or electric shock.	P
10.6	Compliance with the requirement specified in 10.5 may be		P
	accomplished		N
10.7	If any part of a metal spring of a hair clamp of a curling iron or a similar appliance can become loose inside the enclosure of electrical parts as a result of breakage of the spring, the construction shall be such that electrical spacings will not be reduced		N
10.8	Compliance with the requirement specified in 10.7 may be accomplished		N
10.9	The temperature sensor of a temperature controller, a thermostat, a thermal cutoff, or a similar device shall be secured in place.		P
11	**Stability**		**N**
11.1	A floor- or counter-supported appliance shall be constructed such that it will not be overturned when tested in accordance with 37.1.	No overturned.	N
11.2	With regard to 11.1, a hand-supported hair dryer provided with a stand for conversion into a counter-supported hair dryer is to be evaluated as a hand-supported appliance and is not to be subjected to the stability test.		N
11.3	A wax depilatory appliance shall be tested and the results shall be evaluated as described in 37.2, except that the wax may be in any combination of solid and liquid states anticipated during the intended operation of the appliance. Any movable parts or covers are to be in the positions that result in the most adverse conditions of use.		N
12	**Hanging and Mounting Means**		**N**
12.1	A wall-hung or a wall-mounted appliance shall withstand a force as described in 59.1 without evidence of damage to the mounting surface, to the hanging means, to the mounting means, or to the appliance that results in the risk of electric shock, fire, or injury to persons.		N
12.2	A cord-connected appliance that is provided with keyhole slots, notches, hanger holes, or similar feature, for hanging the appliance on a wall		N
12.3	When determining compliance with 12.2, any part of the		N

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Clause	Requirement + Test	Result - Remark	Verdict
	enclosure or barriers that can be removed without the use of tools to gain access to the hanging means is to be removed.		
12.4	A keyhole slot, notch, or hanger hole shall be located so that the supporting screws or similar hardware cannot damage any electrical insulation or reduce spacings to current-carrying parts of the appliance.		N
12.5	A permanently installed wall-mounted appliance shall be provided with the necessary hardware for mounting in accordance with the installation instructions.		N

13	**Supply Connections**		**P**
13.1	Permanently-connected appliances	No permanently-connected appliances.	N
13.1.1	An appliance intended for permanent connection to a power supply, either by being fastened in place, located in A dedicated space, or both, shall have provision for connection of one of the wiring systems that is acceptable for the appliance.		N
13.1.2	The location of a terminal box or compartment in which a power supply connection to a permanently-connected appliance is to be made shall be such that the connection may be readily inspected after the appliance is installed as intended.		N
13.1.3	A terminal compartment intended for the connection of a supply raceway shall be attached to the appliance so as to be prevented from turning.		N
13.2	Wiring terminals	No such Wiring terminals used.	N
13.2.1	An appliance intended for permanent connection to the power supply shall be provided with wiring terminals or leads for connection of supply circuit conductors. Such wiring terminals or leads shall accommodate conductors having an ampacity of not less than 125 percent of the appliance current rating when the load is continuous (3 hours or more), and not less than the appliance current rating when the load is intermittent.		N
13.2.2	For the purpose of these requirements, wiring terminals are considered to be terminals to which power supply or control connections will be made in the field when the appliance is installed.		N
13.2.3	A wiring terminal shall be provided with a soldering lug or with a pressure terminal connector securely fastened in place (for example, firmly bolted or held by a screw).		N
13.2.4	A wiring terminal shall be prevented from turning or shifting in position by means other than friction between surfaces. This may be accomplished by two screws or rivets; by square shoulders or mortices; by a dowel pin, lug, or offset; by a connecting strap or clip fitted into an adjacent part; or by an equivalent means.		N
13.2.5	A wire-binding screw at a wiring terminal shall be no smaller than No. 10 (4.8 mm).		N
13.2.6	A terminal plate tapped for a wire-binding screw shall be of metal not less than 0.050 inch (1.27mm) thick. There shall be two or more full threads in the metal, which may be extruded if necessary to provide the threads.		N

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Clause	Requirement + Test	Result - Remark	Verdict
13.2.7	Upturned lugs or a cupped washer shall be capable of retaining a conductor of the size specified in 13.2.1, but not		N
	smaller than 14 AWG (2.1 mm2), under the head of the screw or the washer.		N
13.2.9	An appliance intended for connection to a grounded power supply conductor and using a:		N
	a) Lampholder or element holder of the Edison screw shell type,		N
	b) Single pole switch, or		N
	c) Single pole automatic control		N
13.2.10	A terminal intended for the connection of a grounded circuit conductor shall be made of, or plated with, a metal substantially white in color and shall be readily distinguishable from the other terminals. If not of such metal, the identification of that terminal shall be clearly shown in some other manner, such as on an attached wiring diagram. A lead intended for the connection of a grounded circuit conductor shall be finished to show a white or gray color and shall be readily distinguishable from the other leads.		N
13.2.11	The free length of a lead inside an outlet box or wiring compartment shall be 6 inches (152.4 mm) or more if the lead is intended for field connection to an external circuit.		N
13.2.12	The surface of an insulated lead intended solely for the connection of an equipment-grounding conductor shall be green, with or without one or more yellow stripes, and no other lead shall be so identified.		N
13.2.13	A wire-binding screw intended for the connection of an equipment grounding conductor shall have a green colored head that is hexagonal shaped, slotted, or both. A pressure wire connector shall be plainly identified as such by being marked″G,″″GR,″″GND,″″Grounding, ″or the like or by a marking on the wiring diagram provided on the appliance. The wire-binding screw or pressure wire connector shall be located so that it is unlikely to be removed during servicing of the appliance.		N
13.2.14	A terminal solely for connection of an equipment grounding conductor shall be capable of securing a conductor of the correct size for that purpose		N
13.3	Cord-connected appliances		P
13.3.1	Cords and plugs		P
13.3.1.1	An appliance shall be provided with a length of flexible cord in accordance with Table 13.1 and an attachment plug for connection to the supply circuit. A coiled cord shall not be used with a floor- or counter-supported appliance where such use would present a risk of burn, fire, electric shock, or injury to persons (for example, the appliance being pulled off a table by the force of the cord). The cord length is measured from the point of cord entry into the enclosure, or into the wiring device at the appliance end of the cord, to		P
	the face of the attachment plug. The length for a coiled cord is to be measured with the cord in an uncoiled position.		P
13.3.1.2	The flexible cord:		P

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Clause	Requirement + Test	Result - Remark	Verdict
	a) May be permanently attached to the appliance or		P
	b) For other than a hand-supported appliance, may be in the form of a detachable power supply cord with means for connection to the appliance.		N
13.3.1.3	The ampacity of the cord (as specified in Table 400.5(A) of the National Electrical Code, ANSI/NFPA 70) and of the plug shall not be less than the current rating of the appliance. The cord and the plug voltage rating shall be at least equal to the rated voltage of the appliance.	No be less than the current rating of the appliance.	P
13.3.1.4	With respect to 13.3.1.3, the voltage rating of a dual voltage appliance is deemed to be that to which the appliance is set when it is shipped from the factory.		P
13.3.1.5	If a dual-voltage appliance is provided with an adapter for connection to an alternate supply source, the adapter shall comply with the applicable requirements in the Standard for Attachment Plugs and Receptacles, UL 498.		P
13.3.1.6	The flexible cord shall be of a type indicated in Table 13.2 or the equivalent.		P
13.3.1.7	The attachment plug of a cord-connected appliance, and the integral blades of a direct plug-in appliance, provided with a 15- or 20-ampere general-use receptacle shall be of the 3-wire grounding type. The attachment plug and the integral blades of all other cord-connected and direct plug-in appliances provided with either a line-connected, single-pole on-off switch or overcurrent protective device, or an Edison-base lampholder shall be polarized or of the grounding type.		N
13.3.1.8	Attachment plugs, appliance couplers, appliance inlets (motor attachment plugs), and appliance (flatiron) plugs, shall comply with the Standard for Attachment Plugs and Receptacles, UL 498. See 16.19 for single and multipole connectors for use in data, signal, control and power applications within and between electrical equipment.		P
13.3.1.9	Female devices (such as appliance couplers, and connectors) that are intended, or that may be used, to interrupt current in the end product, shall be suitably rated		N
	for current interruption of the specific type of load, when evaluated with its mating plug or connector. For example, an appliance coupler that can be used to interrupt the current of a motor load shall have a suitable horsepower rating when tested with its mating plug.		N
13.3.1.10	When a 3-wire grounding-type attachment plug or a 2-wire polarized attachment plug is provided, the attachment plug connections shall comply with Figure 13.2 and the polarity identification of the flexible cord shall comply with Table 13.3.		P
13.3.1.11	Type SPT-2, SVT, or SVTO flexible cord may be used for connecting a pendant-type on-off switch, a temperature control, or both to a table- or floor-supported hair dryer.		N
13.3.1.12	A power supply cord shall not employ conductors smaller than 18 AWG (0.82 mm^2).	No smaller than 0.5 mm^2).	P

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Clause	Requirement + Test	Result - Remark	Verdict
13.3.1.13	A power supply cord with integral fittings shall comply with the requirements in the Standard for Cord Sets and Power-Supply Cords, UL 817, except that it is not Required to be provided with integral overcurrent protection.	No integral fittings.	N
13.3.1.14	Hand-supported appliances provided with detachable power supply cords, including hand-supported Appliances likely to be disconnected while under load, shall not pose a risk of electric shock, fire or injury when mated or disconnected under any orientation or polarity permitted By the construction. The mating connector shall be held securely in place and shall not be allowed to rotate. Compliance is determined by the test of 65.4 and 65.5.		N
13.3.1.15	Appliances provided with a detachable base or stand subject to repeated connection and disconnection during normal use shall not pose a risk of electric shock, fire or injury when mated or disconnected under any orientation or polarity permitted by the construction. The mating connector shall be held securely in place and shall not be allowed to rotate. Compliance is determined by the test of 65.6 and 65.7.		N
13.3.1.16	Appliances provided with a detachable base or stand intended to power a hand-supported appliance which may be disconnected from power during normal use Shall not tip over when the hand-supported portion of the appliance is assembled as intended to the base. Compliance is determined by the stability test of 65.8 and 65.9.		N
13.3.1.17	Locking features provided in accordance with (a) and (b) In the Exception to 13.3.1.2 shall be formed and assembled to have the strength and rigidity required to resist the abuses to which it is able to be subjected. Compliance is determined by the test of 65.10.		N
13.3.1.18	Female contacts and live parts associated with connectors for appliances intended to be disconnected under load during normal use shall not have exposed contacts or terminals accessible to the probe in Figure 13.3.	No any risks.	P
13.3.2	Pin terminals		N
13.3.2.1	When an appliance is provided with pin terminals, the construction of the appliance shall be such that no live part will be exposed to unintentional contact both during and after the placement of the plug on the pins in the intended manner.		N
13.3.2.2	When an appliance is provided with pin terminals, a pin guard is required		N
13.3.2.3	When the pins on the appliance are of American National Standard configuration, the plug used in 13.3.2.2(b) shall consist of an appliance plug in accordance with the Standard for Wiring Devices – Dimensional Requirements, ANSI/NEMA WD6.		N
13.3.2.4	When the pins on the appliance are not of an American National Standard configuration, the plug used in 13.3.2.2(b) shall be the plug supplied with the appliance – 125 volts, 10 amperes, and 250 volts, 5 amperes.		N

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Clause	Requirement + Test	Result - Remark	Verdict
13.3.2.5	When an appliance uses three or more pin terminals intended for use with a plug that covers all the pins, the terminals shall be spaced so that they will not accommodate a flatiron, appliance plug, or cord connector. These pins shall accommodate the plug required for the particular application.		N
13.3.2.6	A pin terminal shall be securely and rigidly mounted and shall be prevented from shifting in position by means other than friction between surfaces.		N
13.3.2.7	The requirement specified in 13.3.2.6 is intended primarily to provide for the maintenance of spacings as specified in 26.1.6, and to maintain required spacings between pin terminals. Under this requirement, consideration is also To be given to the means for locking terminals in position to maintain tightness.		N
13.3.2.8	For a heating appliance, the dimensions of pins and their center-to-center spacings (including the corresponding spacings of the female contacts of general use plugs that these arrangements of pins will accommodate) shall be as indicated in Table 13.4.		N
13.3.2.9	The material on which the pins are mounted, the proximity of any vapor outlet to the terminals, and the direction of the vapor spray shall be such that water shall be prevented from accumulating at the terminal.		N
13.4	Strain relief		P
13.4.1	Strain relief shall be provided such that stress on a Flexible cord will not be transmitted to a terminal, splice, or internal wiring in the appliance or in a fitting (attachment plug, appliance plug, or similar component.		P
13.4.2	If a knot in a flexible cord serves as strain relief, the surface against which the knot bears or with which it contacts shall be free of any projection, sharp edge, burr, fin, results in abrasion of the insulation on the conductors.		P
13.4.3	Insulating bushings serving as strain relief shall comply with the Standard for Insulating Bushings, UL 635. Tests specified in this standard (e.g. Strain Relief Test) may still need to be performed to confirm the combination of the insulating bushing and the supporting part are suitable.		P
13.5	Bushings		P
13.5.1	At a point where a flexible cord passes through an opening in a wall, barrier, or enclosing case, there shall be a bushing or the equivalent that is substantial, reliably secured in place, and that has a smooth, rounded surface against which the cord bears. The bushing or the equivalent is to protect the cord from abrasion damage; it is not intended for strain-relief or flex-relief purposes.		P
13.5.2	In addition to the requirements in 13.5.1, Insulating bushings shall comply with the Standard for Insulating Bushings, UL 635.		P
13.5.3	If the cord hole is in wood, porcelain, phenolic composition, or other nonconducting material, a smooth rounded surface is deemed equivalent to a bushing.		P

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Clause	Requirement + Test	Result - Remark	Verdict
13.5.4	Ceramic materials and some molded compositions are acceptable for insulating bushings. A separate bushing of wood or rubber material (other than in a motor) is not. Vulcanized fiber may be used if the bushing is no less than 1/16 inch (1.6 mm) thick [with a minus tolerance of 1/64 inch (0.4 mm) for manufacturing variations] and if it is formed and secured in place so that it will not be affected adversely by conditions of ordinary moisture.		P
13.5.5	A separate soft rubber, neoprene, or polyvinyl chloride bushing may be used in the frame of a motor or in the enclosure of a capacitor physically attached to a motor (but not elsewhere in an appliance)		P
13.5.6	An insulated metal grommet may be used in place of an insulating bushing if the insulating material used is not less than 1/32 inch (0.8 mm) thick and completely fills the space between the grommet and the metal in which it is mounted.		P
13.6	Direct plug-in appliances	No Direct plug-in appliances.	N
13.6.1	With regard to Figure 13.4, the maximum moment, center of gravity, dimensions, and weight of a direct plug-in appliance shall comply with the requirements specified in (a) – (d). See 13.6.2 and 13.6.3 for symbol definitions and methods of application		N
13.6.2	Definitions for the symbols used in 13.6.1		N
13.6.3	The moment and weight specified in 13.6.1 are to be determined		N
13.6.4	When inserted in a parallel-blade duplex receptacle, any part of an appliance, including a mounting tab, shall not interfere with full insertion of an attachment plug into the adjacent receptacle as illustrated in Figure 13.5.		N
13.6.5	An appliance shall not be provided with a mounting tab unless all the following conditions are met:		N
	a) The appliance is of a type such that semipermanent mounting will not introduce a risk of fire or electric shock;		N
	b) The appliance is intended for use on a 15-ampere, 125-volt receptacle;		N
	c) A screw is provided and constructed so as to secure the mounting tab of the appliance to a parallel-blade duplex receptacle that has a center screw, as shown in Figure 13.5;		N
	d) For an appliance without a grounding pin, the mounting tab is constructed so that the appliance may be mounted to both grounding and nongrounding receptacles; and		N
	e) A marking as specified in 72.7.1 is provided.		N
13.6.6	The enclosure of a direct plug-in appliance shall be capable of being gripped for removal from the receptacle to which it is connected, and the perimeter of the face section from which the blades project shall be no less than 5/16 inch (7.9 mm) from any point on either blade.		N

14	**Live Parts**		**P**
14.1	A current-carrying part shall be of silver, copper, a copper alloy, or equivalent material.		P

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Clause	Requirement + Test	Result - Remark	Verdict
14.2	Plated iron or steel may be used for a current-carrying part	No such part used.	N
14.3	An uninsulated live part shall be secured to the base or mounting surface so that it will be prevented from turning or shifting in position if such motion results in a reduction of spacings below the minimum required values.		N
14.4	Friction between surfaces shall not be used as a means to prevent shifting or turning of an uninsulated live part, but a lock washer applied as intended is acceptable.		N
15	**Reservoirs**		**N**
15.1	If a reservoir is part of an appliance, a live part shall be located or protected so that it will not be subject to dripping if the reservoir does not perform as intended.		N
16	**Internal Wiring**		**P**
16.1	The wiring and connections between parts of an appliance shall be protected or enclosed.		P
16.2	A wireway shall be smooth and entirely free from sharp edges, burrs, fins, moving part similar abrading surfaces that might damage the insulation on the conductors.		P
16.3	A hole in a sheet metal wall through which insulated wires pass shall be provided with a smooth rounded bushing or shall have a smooth, well-rounded surface upon which the wires bear.		N
16.4	A separate foot switch provided with an appliance shall be connected to the appliance by flexible cord no lighter than Type SJ.		N
16.5	Insulated internal wiring (including a grounding conductor) shall consist of a type or types of wire that are acceptable for the application		P
16.6	Internal wiring composed of insulated conductors shall comply with the Standard for Appliance Wiring Material, UL 758.		P
16.7	A splice and connection shall be mechanically secure and shall provide effective electrical contact.		P
16.8	Aluminum conductors, insulated or uninsulated, used as internal wiring, such as for interconnection between current-carrying parts or as motor windings, shall be terminated at each end by a method acceptable for the combination of metals involved at the connection point.		N
16.9	If a wire-binding screw construction or a pressure wire connector is used as a terminating device for aluminum, it shall be required for use with aluminum under the conditions involved (for example, temperature, heat cycling, vibration).		N
16.10	A soldered connection shall be made mechanically secure before being soldered if breaking or loosening of the connection results in a risk of fire, electric shock, or injury to persons		N
16.11	A wire-binding screw or nut shall be provided with a lock-washer if loosening by vibration permits shifting of parts thereby reducing spacings, or otherwise results in a risk of fire, electric shock, or injury to persons. The lock-washer shall be located under the head of a wire-binding screw or under a wire-binding nut.		N

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Clause	Requirement + Test	Result - Remark	Verdict
16.12	An open-end spade lug shall not be used unless additional means (such as upturned ends on the tangs of the lug) are provided to hold the lug in place if the wire-binding screw or nut becomes slightly loosened.		N
16.13	The means of connecting stranded internal wiring to a wire-binding screw shall be such that loose strands of wire are prevented from contacting other live parts not always of the same polarity as the wire and from contacting dead-metal parts. This can be accomplished by using a pressure terminal connector, a soldering lug, a crimped eyelet, or by soldering all strands of the wire together or the equivalent		N
16.14	A splice shall be provided with insulation equivalent to that of the wires involved if spacing between the splice and other metal parts is not permanently maintained.		N
16.15	Insulation consisting of two layers of friction tape, two layers of thermoplastic tape, or one layer of friction tape on top of one layer of rubber tape is acceptable on a splice. In determining whether splice insulation consisting of coated fabric, thermoplastic, or other type Of tubing is acceptable, consideration is to be given to such factors as dielectric properties, heat- and moisture-resistant characteristics, and similar criteria. Thermoplastic tape wrapped over a sharp edge shall not be used.		N
16.16	Quick-connect type wire connectors shall be suitable for the wire size, type (solid or stranded), conductor material (copper or aluminum) and the number of conductors terminated. If insulated, they shall be rated		N
	for the voltage and temperature of the intended use. They shall be applied per the installation instructions of the wire connector manufacturer.		N
16.17	Quick-connect terminals, both connectors and tabs, for use with one or two 22 – 10 AWG copper conductors, having nominal widths of 2.8, 3.2, 4.8, 5.2, and 6.3 mm (0.110, 0.125, 0.187, 0.205, and 0.250 inch), intended for internal wiring connections in appliances, or for the field termination of conductors to the appliance, shall comply with the Standard for Electrical Quick-Connect Terminals, UL 310.		N
16.18	Wire connectors shall comply with the Standard for Wire Connectors, UL 486A-486B.		N
16.19	Splicing wire connectors shall comply with the Standard for Splicing Wire Connectors, UL 486C		N
16.20	Single and multipole connectors for use in data, signal, control and power applications within and between electrical equipment, and that are intended for factory assembly to copper or copper alloy conductors, or for factory assembly to printed wiring boards, shall comply with the Standard for Component Connectors for Use in Data, Signal, Control and Power Applications, UL 1977.		P
16.21	Multi-pole splicing wire connectors that are intended to facilitate the connection of hard-wired utilization equipment to the branch-circuit conductors of buildings shall comply with the Standard for Insulated Multi-Pole Splicing Wire Connectors, UL 2459.		N

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Clause	Requirement + Test	Result - Remark	Verdict
16.22	Equipment wiring terminals for use with all alloys of copper, aluminum, or copper-clad aluminum conductors, shall comply with the Standard for Equipment Wiring Terminals for Use with Aluminum and/or Copper Conductors, UL 486E.		P
16.23	Terminal blocks shall comply with the Standard for Terminal Blocks, UL 1059, and, if applicable, be suitably rated for field wiring		N
17	**Heating Element**		**P**
17.1	A heating element shall be supported in a reliable manner and shall be protected against mechanical damage and contact with outside objects.		P
17.2	In determining whether a heating element is reliably supported, consideration is to be given to sagging, loosening, and other adverse conditions resulting from continuous heating.		P
17.3	An appliance in which the heating element is designed for operation only in an air stream shall be wired or controlled so that the element is capable of operation only when under the cooling effect of the air stream.		P
17.4	A sheathed element, open-wire heating element or the		P
	like shall be judged under the applicable requirements of this Standard.		P
17.5	Insulated heating wire shall comply with the Standard for Appliance Wiring Material, UL 758.		P
17.6	Thermistor-type heaters (e.g. PTC or NTC heaters) shall comply with the Standard for Thermistor-Type Devices, UL 1434.	PTC	P
18	**Electrical Insulation**		**P**
18.1	General		P
18.1.1	An insulating washer, bushing, or similar part that is an integral part of an appliance, and a base or support for the mounting of a current-carrying part, shall be of a moisture-resistant material that will not be adversely affected by the temperatures to which it will be subjected under conditions of intended use. Molded parts shall be constructed so that they will have strength and rigidity to withstand the stresses of intended service.		P
18.1.2	Insulating material is to be evaluated with respect to its acceptability for the particular application.		N
18.1.3	In the mounting or supporting of a small fragile insulating part, a screw or other fastening is not to be so tight as to result in cracking or breaking with expansion and contraction. Such a part shall be slightly loose.		N
18.1.4	A small molded part, such as a brush cap, shall be constructed so that it will have the strength and rigidity to withstand stresses during intended use.		P
18.1.5	Insulating material on which the opposite polarity fixed contacts of a hand-held hair dryer power “on-off” slide switch are mounted shall have a comparative tracking index (CTI) rating of 2 or better, and a flammability rating of V-1 or better.		N

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Clause	Requirement + Test	Result - Remark	Verdict
18.1.6	Insulating tape shall comply with the Standard for Polyvinyl Chloride, Polyethylene, and Rubber Insulating Tape, UL 510.		N
18.1.7	Insulation sleeving shall comply with the Standard for Coated Electrical Sleeving, UL 1441		P
18.1.8	Insulation tubing shall comply with the Standard for Extruded Insulating Tubing, UL 224.		N
18.1.9	A printed-wiring board shall comply with the requirements in the Standard for Printed-Wiring Boards, UL 796.		N
18.1.10	Unless otherwise specified, the flammability class and temperature rating shall be that specified for insulating materials in the Standard for Polymeric Materials – Use in Electrical Equipment Evaluation, UL 746C.		P
18.2	Film-coated wire (magnet wire)		N
18.2.1	The component requirements for film coated wire and class 105 (A) insulation systems are not specified.		N
18.2.2	Film coated wire in intimate combination with one or more insulators, or the magnet wire of induction heating coil, incorporated with an insulation system rated Class 120 (E) or higher, shall comply with the magnet wire requirements in the Standard for Systems of Insulating Materials – General, UL 1446 and shall have a suitable temperature class.		N
19	**Thermal Insulation**		**P**
19.1	Combustible thermal and electrically conductive insulation shall not contact an uninsulated live part.		P
19.2	Mineral wool thermal insulation that contains conductive impurities in the form of slag shall not come into contact with any uninsulated live part.		P
19.3	Thermal insulation shall be rated for the temperature to which it is exposed when tested under the conditions described in 44.1.1.		P
20	**Overcurrent Protection**		**P**
20.1	If overcurrent conditions are likely to occur, the appliance shall be provided with a circuit breaker or fuse.		P
20.2	Overcurrent protection at not more than 20 amperes shall be provided by means of a circuit breaker or fuse in the appliance for each general use receptacle circuit And each lampholder circuit in the appliance, unless the appliance would be correctly connected to a branch circuit rated at 20 amperes or less.	No more than 20 amperes.	P
20.3	The overcurrent protection specified in 20.2 shall be of a type rated for branch circuit protection		P
20.4	A fuseholder or circuit breaker provided as a part of an appliance shall be of a type rated for the particular application and shall not be accessible from outside the appliance without opening a door or cover. A fuseholder for a plug fuse shall be constructed and installed so that an uninsulated live part other than the screw shell will not be exposed to contact by persons removing or replacing a fuse.		N
20.5	For other than a hand-supported appliance, if the handle of a circuit breaker is operated vertically rather than		N

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Clause	Requirement + Test	Result - Remark	Verdict

	rotationally or horizontally, the up position of the handle shall be the on position.		

21	**Thermal Cutoffs (Fusible Links)**		**N**
21.1	If an appliance is provided with a thermal cutoff, the cutoff shall open the circuit in the intended manner without causing the short circuiting of live parts and without		N
	causing live parts to become grounded to the enclosure. This determination is to be made in accordance with the test requirements specified in the Test of Thermal cutoffs (Fusible Links), Section 55.		N
21.2	A thermal cutoff shall comply with the Standard for Thermal-Links – Requirements and Applications Guide, UL 60691.		N

22	**Lampholders and Receptacles**	No Lampholders and Receptacles used.	**N**
22.1	Lampholders and indicating lamps integral with lampholder shall comply with the Standard for Lampholders, UL 496. A female screw shell used as a holder for a heating element shall be of copper or of copper alloy and shall be plated with nickel or an equivalent oxidation-resistant metal		N
22.2	The circuit conductor of a power supply cord that is intended to be grounded shall have the following items connected to it:		N
	a) The screw shell of an Edison-base lampholder and		N
	b) The terminal or lead of a receptacle intended to be grounded.		N
22.3	An Edison-base lampholder shall not be used in an appliance rated over 150 volts.		N
22.4	In determining compliance with the Exception to 22.3, the probe shown in Figure 8.1 shall be used as described in 8.5.3.		N
22.5	An Edison-base lampholder in an appliance rated 150 volts or less shall be constructed or installed so that an uninsulated live part other than the screw shell will not be exposed to contact by a person removing or replacing a lamp during intended service.		N
22.6	A 15- or 20-ampere attachment plug receptacle intended for general use in an appliance shall be of the grounding type. The grounding contact of the receptacle shall be electrically connected to dead metal that will be grounded when the appliance is in use.		N
22.7	Attachment plug receptacle shall comply with the Standard for Attachment Plugs and Receptacles, UL 498.		N
22.8	The face of a receptacle that is less than 5/8 inch (15.9 mm) wide or 7/8 inch (22.2 mm) long shall project a minimum of 0.015 inch (0.38 mm) and a maximum of 3/16 inch (4.8 mm) from the part of the receptacle-mounting surface that is within a rectangle 5/8 inch wide and 7/8 inch long, the rectangle being symmetrically located about the receptacle contacts.		N
22.9	An appliance provided with one or more general use receptacles shall not be equipped with a flexible cord not smaller than 16 AWG (1.3 mm2).		N

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Clause	Requirement + Test	Result - Remark	Verdict

22.10	When the branch circuit over current protection will be inadequate for any general use receptacle or receptacles provided as part of an appliance, over current protection for the receptacle or receptacles shall be provided as part of the appliance		N

23	**Switches**		**P**
23.1	General		P
23.1.1	An appliance having any driven moving part, which by function could cause entrapment of hair, body parts, clothing or the like, shall be provided with a main on-off switch. Appliances in this group include, but are not limited to, hair dryers, hair untanglers, and the like.		N
23.1.2	A switch, as required in 23.1.1, shall be located so that it can be operated by the user to turn off the appliance.		P
23.1.3	A switch shall be acceptable for the particular application and shall have a current and voltage rating no less than that of the circuit (load) it controls. See 18.1.5 for electrical insulation of slide switch.		P
23.1.4	Manually operated snap-switches shall comply with one of the following, as applicable:		N
	a) Standard for Switches for Appliances – Part 1: General Requirements, UL 61058-1;		N
	b)Deleted;		N
	c) Standard for General-Use Snap Switches, UL 20; or		N
	d) Standard for Nonindustrial Photoelectric Switches for Lighting Control, UL 773A.		N
23.1.5	A clock-operated switch, in which the switching contacts are actuated by a clock-work, by a gear-train, by electrically-wound spring motors, by electric clock-type motors, or by equivalent arrangements		N
23.1.6	A timer or time switch, incorporating electronic timing circuits or switching circuits, with or without separable contacts, shall comply with the requirements for an operating control with Type 1 action for 6000 cycles of operation, or as a manual control for 5000 cycles of operation, in accordance with the following:		N
	a) The Standard for Automatic Electrical Controls for Household and Similar Use; Part 1: General Requirements, UL 60730-1; and the Standard for Automatic Electrical Controls for Household and Similar Use; Part 2: Particular Requirements for Timers and Time Switches, UL 60730-2-7 or		N
	b) The Standard for Solid-State Controls for Appliances, UL 244A		N
23.1.7	A manually operated, line-connected, single-pole switch		N
	for appliance on-off operation shall not be connected to the conductor of the power supply cord or circuit intended to be grounded. Table 13.3 specifies the identification of the power supply cord conductor intended to be grounded.		
23.1.8	A switch that is subjected to a temperature of more than 65°C (149°F) shall be evaluated with respect to the temperature limitations of the materials used.		N

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Clause	Requirement + Test	Result - Remark	Verdict
23.1.9	A switch shall be located or protected so that it will not be subjected to mechanical damage during use.		P
23.1.10	A switch, as required in 23.1.1, shall have a plainly marked″off″position. The use of a symbol alone, such as the symbol″O,″shall not be used to denote the off position. The switch position marking need not be an integral part of the switch itself.		N
23.1.11	A hand-supported hair-drying appliance is not intended to be immersible and shall not be so marked.		N
23.1.12	A maintained contact switch for a hand-supported hair dryer shall be subjected to a 6,000-cycle switch endurance test. A momentary contact switch that is likely to be operated several times during each use of a hand-supported hair dryer, such as the on-off switch, shall be subjected to a 30,000-cycle switch endurance test. A momentary-contact switch that is not likely to be operated several times during each use of a hand-supported hair dryer, such as a switch used to provide low-velocity, cool air for setting a curl (a″cool shot″ switch), shall be subjected to 6,000 cycles of the switch endurance test. The tests, when required, are to be conducted in accordance with the Standard for Switches for Appliances – Part 1: General Requirements, UL 61058-1.		N
23.2	Dual-voltage selector		N
23.2.1	The construction of the supply circuit voltage selector shall be such that the supply circuit voltage setting cannot be changed without the use of a tool (a coin, screwdriver, or the like is considered to be a tool for the purpose of this requirement).		N
23.2.2	If the appliance is constructed so that the supply circuit voltage selector setting can be changed, the action of changing the voltage selector setting shall also change the supply circuit voltage indication.		N
23.2.3	An appliance that can be set to different rated supply circuit voltages shall be provided with the statement required in 74.7(k)(13 – 14).		N
24	**Automatic Controls and Control Circuits**		**P**
24.1	General		P
24.1.1	The operation of an automatic control device in an appliance shall disconnect the element or elements it controls from all ungrounded conductors of the supply circuit.		P
24.1.2	Breakdown of a temperature control in a hand-supported hair dryer shall not result in a risk of fire, electric shock, or injury to persons as determined in accordance with 46.4.2.4 and 46.4.3.3. A limit control that operates to interrupt all heater and motor circuits and end the test shall comply with the requirements specified in Thermal Cutoffs (Fusible Links), Section 21.		N
24.1.3	The overload and endurance tests of a temperature controller consisting of a temperature sensor and the associated control circuit for an appliance having a preheat cycle shall be conducted in the appliance, or under conditions representative of those in the appliance, as described in the Test of Automatic		N

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Clause	Requirement + Test	Result - Remark	Verdict
	Controls, Section 54. See 24.2.4.		
24.1.4	A temperature controller that controls the duration of a preheat cycle by a timing circuit or by an equivalent means without using a temperature sensor is considered to be a temperature-regulating control and shall comply with the overload and endurance requirements specified in the Test of Automatic Controls, Section 54.		N
24.1.5	Auxiliary controls shall be evaluated in accordance with the applicable requirements of this standard and the parameters in Controls – End Product Test Parameters, Section 25 unless otherwise specified in this standard. See 24.1.12.		N
24.1.6	Operating (regulating) controls shall be evaluated in accordance with the applicable component standard requirements specified in 24.2, if applicable, and the parameters in Controls – End Product Test Parameters, Section 25, unless otherwise specified in this standard. See 24.1.12.		P
24.1.7	Electronic operating controls that rely upon software for the normal operation of the end product where deviation or drift of the control may result in a risk of fire, electric shock, or injury to persons		N
24.1.8	Protective (limiting) controls shall be evaluated in accordance with the applicable component standard requirements specified in 24.2 and if applicable, the parameters in Controls – End Product Test Parameters, Section 25, unless otherwise specified in this standard.		N
24.1.9	Electronic protective controls that do not rely upon software as a protective component		N
24.1.10	Electronic protective controls that rely upon software as a protective component		N
24.1.11	If a single malfunction or breakdown of an electronic component, located in electronic operating control, results in increased risk of injury to persons, such as a loss of OFF control or unexpected operation, this control shall comply with the applicable requirements specified in 24.1.7 – 24.1.10. See 25.4.		N
24.1.12	An electronic, auxiliary or operating control (e.g. a non-protective control), the failure of which would not increase the risk of fire, electric shock, or injury to persons (i.e. burn injury), is not required to comply with the requirements in 24.1.6 – 24.1.11, and is only required to be subjected to the applicable requirements of this standard.		N
24.2	Electromechanical and electronic controls		N
24.2.1	A temperature control shall comply with one of the following:		N
	a) The Standard for Temperature-Indicating and -Regulating Equipment, UL 873 or		N
	b) The Standard for Automatic Electrical Controls for Household and Similar Use; Part 1: General Requirements, UL 60730-1; and the Standard for Automatic Electrical Controls for Household and Similar Use; Part 2: Particular Requirements for Temperature Sensing Controls, UL 60730-2-9.		N
24.2.2	A temperature control as noted in 5.36 installed in a hand-supported hair dryer shall operate at not more than 8.3°C		N

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Clause	Requirement + Test	Result - Remark	Verdict
	(15°F) above or below its rated operating temperature. Compliance is determined by subjecting the control, a sub-assembly including the control, or the complete appliance to the appropriate temperatures in an air oven.		
24.2.3	In a wax depilatory appliance, an automatic-reset temperature control shall be a calibrated control endurance tested for at least 6,000 cycles of operation and shall comply with all other requirements applicable to limit controls in the Standard for Limit Controls, UL 353, or the requirements applicable to temperature-limiting controls in the Standard for Temperature-Indicating and -Regulating Equipment, UL 873. The calibration requirements shall be as specified for water-heater limit controls in UL 353 or water-heater temperature-limiting controls in UL 873. Compliance with the Standard for Automatic Electrical Controls for Household and Similar Use, Part 1: General Requirements, UL 60730-1, and/or the applicable Part 2 standard from the UL 60730 series fulfills the UL 873 requirements.		N
24.2.4	A temperature sensing positive temperature coefficient (PTC) or a negative temperature coefficient (NTC) thermistor, that performs the same function as an operating or protective contro		P
25	**Controls – End Product Test Parameters**		**N**
25.1	General		N
25.1.1	Spacings of controls shall comply with the electrical spacing, or clearances and clearance distance requirements of the applicable control standard as determined in Spacings, Section 26.		N
25.1.2	Where reference is made to declared deviation and drift, this indicates the manufacturer's declaration of the control's tolerance before and after certain conditioning tests.		N
25.2	Auxiliary controls		N
25.2.1	Auxiliary controls shall not introduce a risk of electric shock, fire, or personal injury		N
25.2.2	Auxiliary controls shall comply with the requirements of this standard.		N
25.3	Electronic Operating controls (regulating controls)		N
25.3.1	The following test parameters shall be among the items considered when judging the acceptability of an operating control investigated using the Standard for Automatic Electrical Controls for Household and Similar Use, Part 1: General Requirements, UL 60730-1		N
25.3.2	The following test parameters shall be among the items considered when judging the acceptability of an operating control investigated using other than the Standard for Automatic Electrical Controls for Household and Similar Use, Part 1: General Requirements, UL 60730-1		N
25.4	Electronic Protective controls (limiting controls)		N
25.4.1	An electronic control that performs a protective function shall comply with the applicable requirements in Automatic		N

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Clause	Requirement + Test	Result - Remark	Verdict
	Controls and Control Circuits, Section 24 while tested using the parameters in this Section.		
25.4.2	The following test parameters shall be among the items considered when judging the acceptability of an electronic protective control investigated using the Standard for Automatic Electrical Controls for Household and Similar Use, Part 1: General Requirements, UL 60730-1		N
25.4.3	The test parameters and conditions used in the investigation of the circuit covered by 25.4.1 shall be as specified in the Standard for Tests for Safety-Related Controls Employing Solid-State Devices, UL 991		N
25.4.4	Unless otherwise specified in this Standard, protective controls shall be evaluated for 100,000 cycles for Type 2 devices, and 6,000 cycles for Type 1 devices, with rated current.		N
26	**Spacings**		**P**
26.1	General		P
26.1.1	All uninsulated live parts connected to different circuits – line voltage, low voltage (Class 2), or limited energy primary – and separated electrically by insulation or impedance shall be spaced from one another as though they were parts of opposite polarity and shall be judged on the basis of the highest voltage involved		P
26.1.2	The spacing between uninsulated live parts of opposite polarity and between such parts and dead metal that may be grounded in service is not specified for parts of circuits that are classified as low-voltage (Class 2) circuits.		P
26.1.3	The spacing between uninsulated live parts within a limited energy primary circuit is not specified		N
26.1.4	The spacing between uninsulated live parts of a limited-energy primary circuit and dead metal that may be contacted by persons, or that may become grounded in service, is as specified in 26.1.6.		N
26.1.5	With respect to 26.1.4, an entire component shall be evaluated as live part if any dead metal of the component is isolated from a live part by an insulation system or by a spacing that is inadequate for the line voltage involved.		P
26.1.6	There shall be a spacing of not less than 1/16 inch (1.6 mm) between uninsulated line voltage parts of opposite polarity, and between an uninsulated line-voltage part and a dead-metal part that might be exposed to contact by persons during operation of the appliance or that might be grounded.		P
26.1.7	An insulating lining or barrier of fiber or similar material shall be so located or of such material that it will not be affected adversely by arcing. If the lining or barrier is used instead of an air spacing, the material shall not be less than 1/32 inch (0.8 mm) thick.		P
26.2	Spacings on printed wiring boards		p
26.2.1	As an alternative to the spacing requirements in 26.1.6, a printed wiring board with spacings between opposite polarity		N

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Clause	Requirement + Test	Result - Remark	Verdict
	circuits (other than a low-voltage circuit) less Than those required is acceptable		
26.2.2	When conducting evaluations in accordance with the requirements in the Standard for Insulation Coordination Including Clearances and Creepage Distances for Electrical Equipment, UL 840		N
26.2.3	In order to apply Clearance B (controlled overvoltage) clearances, control of overvoltage shall be achieved by providing an overvoltage device or system as an integral part of the product. This voltage limiting device or system shall comply with the Standard for Surge Protective Devices, UL 1449.		N
27	**Grounding**	No Grounding used.	**N**
27.1	All permanently connected appliances shall have provision for the grounding of all exposed metal parts that are likely to become energized.		N
27.2	An appliance marked as double insulated shall not be provided with a means for grounding.		N
27.3	If a grounding means is provided on the appliance, whether required or not, all exposed dead-metal parts and all dead-metal parts within the enclosure that are exposed to contact during any servicing operation and that are likely to become energized shall be reliably connected to the grounding means		N
27.4	An equipment grounding conductor of a flexible cord		N
28	**Motors**		**P**
28.1	Construction		P
28.1.1	A motor provided as part of an appliance shall be capable of handling the load it is intended to drive without introducing a risk of fire, electric shock, or injury to persons.		P
28.1.2	A motor winding shall be constructed so as to resist the absorption of moisture.		P
28.1.3	With reference to the requirement specified in 28.1.2, film-coated wire is not required to be additionally treated to prevent absorption of moisture. Fiber slot liners, cloth coil wrap, and similar moisture-absorptive materials shall be provided with impregnation or otherwise treated to prevent moisture absorption.		P
28.1.4	A brush cap, accessible from outside an enclosure of a portable appliance that prevents contact with a live part at a potential of more than 30 volts rms (42.4 volts peak) to any other part or to ground, shall be fastened in place so that removal cannot be accomplished by an ordinary tool used in the intended manner. Wrenches, pliers, and flat-blade or cross-blade screwdrivers are deemed to be ordinary tools.		P
28.2	Brush wear-out		P
28.2.1	A brush-holder assembly shall be constructed so that when a brush is worn out (no longer capable of performing its function), the brush, spring, and other parts of the assembly will be retained to the degree necessary to reduce the likelihood of:		P

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Clause	Requirement + Test	Result - Remark	Verdict
	a) Accessible dead-metal parts becoming energized and		P
	b) Live parts becoming accessible.		N
28.2.2	With reference to the requirement in 28.2.1, the parts of a brush holder assembly are considered to be acceptably retained		N
28.2.3	A motor control device not having a horsepower rating equivalent to the motor it controls, shall be capable of performing effectively when subjected to an overload test as specified in the Motor Control Overload Test, Section 56.		P
28.3	Overload protection		P
28.3.1	Except as indicated in 28.3.8, the following appliances in whicha1hporsmaller motor is used shall incorporate thermal or overload protection that prevents the motor from attaining excessive temperatures under any operating conditio		P
28.3.2	An appliance intended to be automatically or remotely controlled, and employing a motor rated at more than 1 hp, shall incorporate thermal or overcurrent protection.		N
28.3.3	Fuses shall not be used as motor-overload-protective devices unless the motor is protected by the largest size of fuse that can be inserted in the fuseholder.		P
28.3.4	Thermal protection devices integral with the motor shall comply with one of the following:		N
	a)Deleted;		N
	b) The Standard for Thermally Protected Motors, UL 1004-3; or		N
	c) The Standard for Automatic Electrical Controls for Household and Similar Use, Part 1: General Requirements, UL 60730-1; and the Standard for Automatic Electrical Controls for Household and Similar Use; Part 2 Particular Requirements for Thermal Motor Protectors, UL 60730-2-2; in conjunction with the Standard for Thermally Protected Motors, UL 1004-3 (to evaluate the motor-protector combination).		N
28.3.5	Impedance protection shall comply with the Standard for Impedance Protected Motors, UL 1004-2		N
28.3.6	Electronic protection integral to the motor shall comply with the Standard for ElectronicallyProtected Motors, UL 1004-7.		N
28.3.7	Electronically protected motor circuits		N
28.3.8	Motors are not required to comply with the overload protection requirement		N
28.3.9	Running overload protection is not required		N
28.3.10	An overload-protective device that complies with the National Electrical Code, ANSI/NFPA 70, is determined to comply with 28.3.1. This overload-protective device shall be responsive to motor current and rated or set as specified in Column A of Table 430-72(b) of the NEC.		N
	When the rating of the motor-running overload protection determined in accordance with the foregoing does not correspond to a standard size or rating of a fuse, nonadjustable circuit breaker, thermal cutout, thermal relay, or heating element of a thermal-trip motor switch, the next higher size, rating, or setting is not prohibited from being		N

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	used, and shall not be more than that specified in Column B of Table 430-72(b) of the NEC. For a multi-speed motor, each winding connection is to be evaluated separately.		
28.3.11	The functioning of an overload protective device, whether or not such a device is required, shall not result in a risk of fire, electric shock, or injury to persons. Overload devices used for running overload protection, other than those that are inherent in a motor, shall be located in each ungrounded conductor of a supply system.		N
28.3.12	Motor-overload protection in which contacts control a relay coil in a motor starter shall comply with the requirements of 28.3.1.		N
28.3.13	Fuses used in motor-overload-protective devices shall be configured so that the motor is investigated with the largest size of fuse that is capable of being inserted in the fuseholder.		N
28.4	Insulation systems		P
28.4.1	Class A insulation systems shall consist of a combination of magnet wire and major component insulation materials evaluated and found to operate as intended in its end use. Thermoset materials and materials specified in Table 28.1 at the thicknesses specified are permitted to be used without further evaluation.		N
28.4.1	Class A insulation systems shall consist of a combination of magnet wire and major component insulation materials evaluated and found to operate as intended in its end use. Thermoset materials and materials specified in Table 28.1 at the thicknesses specified are permitted to be used without further evaluation.		N
28.4.2	For Class A insulation systems employing other materials or thinner materials than those indicated in Table 28.1 or a combination of materials, the materials, whether polymeric or not polymeric (treated cloth, for example), shall comply with the requirements in 28.4.3		N
28.4.3	A polymeric material employed in a Class 105 (A) insulation system that isolates the windings from dead metal parts shall be unfilled or glass-reinforced nylon, polycarbonate, polybutylene terephthalate, polyethylene terephthalate, phenolic or acetal, and shall have a relative or generic thermal index for electrical properties of 105°C (221°F) minimum. Leads shall be rated 90°C (194°F) minimum. Motors employing thermoplastic materials shall be subjected to the tests in Thermoplastic Motor Insulation Systems, Section 66.		N
28.4.4	Materials used in an insulation system that operates above Class 105 (A) temperatures shall comply with the Standard for Systems of Insulating Materials – General, UL 1446.		N
28.4.5	All insulation systems employing integral ground insulation shall comply with the requirements specified in the Standard for Systems of Insulating Materials – General, UL 1446		P

29	**Transformers**		**N**
29.1	General-purpose transformers shall comply with the	No such transformers	N

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	Standard for Low Voltage Transformers – Part 1: General Requirements, UL 5085-1 and the Standard for Low Voltage Transformers – Part 2: General Purpose Transformers, UL 5085-2		
29.2	Class 2 and Class 3 transformers shall comply with the Standard for Low Voltage Transformers – Part 1: General Requirements, UL 5085-1 and the Standard for Low Voltage Transformers – Part 3: Class 2 and Class 3 Transformers, UL 5085-3.		N
30	**Batteries and Battery Chargers**		**N**
30.1	A lithium ion (Li-On) single cell battery shall comply with the requirements for secondary lithium cells in the Standard for Lithium Batteries, UL 1642.		N
30.2	Rechargeable nickel cadmium (Ni-Cad) cells and battery packs shall comply with the applicable construction and performance requirements of this standard.		N
30.3	Rechargeable nickel metal-hydride (Ni-MH) battery cells and packs shall comply with the applicable construction and performance requirements of this standard, or the applicable requirements for secondary cells or battery packs in the Standard for Household and Commercial Batteries, UL 2054.		N
30.4	Primary batteries (non-rechargeable) that comply with the requirements of the relevant UL Standard and of 2.4 are considered to fulfill the requirements of this standard.		N
30.5	A Class 2 battery charger		N
31	**Capacitors**		**P**
31.1	A capacitor provided as a part of a capacitor motor and a capacitor connected across the line, such as a capacitor for radio-interference elimination or power-factor correction, shall be housed within an enclosure or container that shall protect the plates against mechanical damage and prevent the emission of flame or molten material resulting from malfunction of the capacitor. The container shall be of metal providing strength and protection not less than that of uncoated steel having a thickness of 0.020 inch (0.51 mm). Sheet metal having a thickness less than 0.026 inch (0.66 mm) shall not be used. The motor starting or running capacitor shall comply with the Standard for Capacitors, UL 810. The across the line capacitor shall comply with the applicable requirements in the Standard for Fixed Capacitors for Use in Electronic Equipment – Part 14: Sectional Specification: Fixed Capacitors for Electromagnetic Interference Suppression and Connection to the Supply Mains, UL 60384-14.		P
31.2	If a capacitor that is not a part of a capacitor motor or a capacitor-start motor is connected in an appliance that is intended to be automatically or remotely-controlled so that malfunction or breakdown of the capacitor would result in a risk of fire, electric shock, or injury to persons, thermal or overcurrent protection shall be provided in the		P

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	appliance to reduce such a risk.		
31.3	Capacitors, connected line-to-line or line-to-ground, shall comply with the requirements in the Standard for Fixed Capacitors for Use in Electronic Equipment – Part 14: Sectional Specification: Fixed Capacitors for Electromagnetic Interference Suppression and Connection to the Supply Mains, UL 60384-14. A capacitor connected from one side of the line to the frame or enclosure of an appliance shall have a capacitance rating of not more than 0.10 μF.		P
31.4	The voltage rating of a capacitor other than a motor-starting or motor-running capacitor shall equal or exceed the maximum steady-state potential to which the capacitor is subjected during operation of the unit at the rated voltage.		P
31.5	Under both normal and abnormal conditions of use, a capacitor employing a liquid dielectric medium more combustible than askarel shall not cause a risk of electric shock or fire and shall be protected against expulsion of the dielectric medium.		P

32	**Light Sources and Associated Components**		N
32.1	Lighting ballasts shall comply with the Standard for		N
	Fluorescent-Lamp Ballasts, UL 935, or the Standard for High-Intensity Discharge Lamp Ballasts, UL 1029. Ballasts forming part of a luminaire that complies with an appropriate luminaire standard are considered to fulfill this requirement.		N
32.2	Light emitting diode (LED) light sources shall comply with the Standard for Light Emitting Diode (LED) Equipment For Use In Lighting Products, UL 8750. LED light sources forming part of a luminaire that complies with an appropriate luminaire standard are considered to fulfill this requirement.		N

33	**Ionization Circuits**		**N**
33.1	Grooming appliances which employ ionization technology shall comply with 33.2 and 33.3.		N
33.2	The high voltage power supply used in the ionizer shall be evaluated to the applicable construction and component requirements for power supplies contained in the Standard for Electrostatic Air Cleaners, UL 867.		N
33.3	The high voltage pins (electrodes) of ionizer shall not be accessible per 8.5.		N
33.4	A grooming appliance employing ionization circuitry shall not produce a concentration of ozone exceeding 0.05 parts per million by volume when tested as described in Ozone Test, Section 67.		N
33A	Button or Coin Cell Batteries of Lithium Technologies		N
33A.1	To reduce the risk of injury due to battery ingestion by small children, the battery compartment of an appliance or any accessory, such as a wireless control, incorporating one or more coin cell batteries of lithium technologies shall comply with the Standard for Products Incorporating Button or Coin Cell Batteries of Lithium Technologies, UL 4200A,		N

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	if the appliance or any accessory is intended for use with one or more single cell batteries having a diameter of 1.25 inches (32 mm) maximum with a diameter greater than its height.		

PERFORMANCE			
34	**General**		**P**
34.1	The tests described in Sections 38 – 46 shall be conducted in that order on the same samples.		P
34.2	A simulated head used for temperature testing is to consist of a foamed plastic wig form, approximately 21-1/2 inches (546 mm) in circumference, closely wrapped with two layers of cheesecloth. Pieces of black (exposed and developed) cellulose acetate photographic film to represent hair-holding devices are to be attached to the top and sides.		P
34.3	Wherever cheesecloth is specified in connection with either a temperature test or an abnormal test, the cloth is to be bleached cheesecloth 36 inches (914 mm) wide, running 14 – 15 yards per pound mass (approximately 28 – 30 m/kg mass), and having what is known in the trade as a″count of 32 x 28,″which means that for any square inch there are 32 threads in one direction and 28 threads in the other direction (for any square centimeter there are 13 threads in one direction and 11 threads in the other direction).		P
34.4	For the purpose of these requirements, a primary temperature limiting control in an appliance that has two different temperature limiting controls is the control that is intended to operate before the second control operates. The second control, termed the backup temperature limiting control, is intended to operate in the event of malfunction of the primary control.		N
34.5	Wherever a hardwood surface is specified in connection with a test, the hardwood surface is to consist of a layer of tongue-and-groove oak flooring mounted on two layers of nominal 3/4 inch (19.1 mm) plywood. The oak flooring is to be nominally 3/4 inch thick [actual size 3/4 by 2-1/4 inch (19.1 by 57.2 mm)]. The assembly is to rest on a concrete floor or an equivalent nonresilient floor during the test.		N

35	**Strength of Enclosure Test**		**P**
35.1	A 5-pound (22.2-N) force shall be applied by means of the flat end of a circular steel rod that is 1/4 inch (6.4 mm) in diameter and 5 inches (127 mm) long for 1 minute to any part of the area described in 8.5.8. The rod is to be vertical, and the appliance may be oriented in any position relative to the rod before the force is applied.		P
	a) During the test, the rod does not contact an uninsulated live part; and		P
	b) After the test, the construction is in compliance with 8.5.3, 8.5.4, and 26.1.6.		P
35.2	With reference to 35.1, the test is to be conducted on a guard such as a screen, which is located under an		N

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Clause	Requirement + Test	Result - Remark	Verdict
	opening in an enclosure, through the opening in the enclosure only		
36	**Tip-Over Test**		**N**
36.1	Three samples of an appliance as described in 9.2.1 shall be tested and each sample is to be tested three times. Each sample of the appliance is to be placed on a horizontal surface of laminated thermosetting counter-top-type material. The appliances are to be oriented in a position that is likely to occur during intended use, and are to contain whatever combination of separable components and liquid that results in the most adverse condition for this test.		N
36.2	For an appliance with a capacity of 32 fluid ounces (947 mL) or less, the sample is to be tilted to determine its critical angle of balance (the angle at which the sample will tip over due only to the force of gravity).		N
36.3	For an appliance with a capacity of greater than 32 fluid ounces (947 mL), the sample is to be tipped over. The results are acceptable if the lid remains in place.		N
37	**Stability Test**		**N**
37.1	In accordance with 11.1, an appliance shall be placed on a supporting surface that is inclined at a 10-degree angle from the horizontal. The appliance is to be turned to the position most likely to cause tipping. Any adjustable or movable part that will affect the location of the center of gravity of the appliance is to be placed in the position most likely to contribute to tipping. If the appliance is on castors, blocks are to be placed in front of them to prevent the appliance from moving down the incline. The results are acceptable if the appliance remains stable on the tilted surface.		N
37.2	An appliance of the type specified in 9.2.1 shall be placed on a plane inclined at an angle of 15 degrees to the horizontal. The appliance shall be positioned and loaded with whatever combination of separable components (strainers, cups, and similar parts) and liquid that results in the maximum tendency to overturn under conditions of intended use. The appliance shall be prevented from sliding on the inclined surface. The result is acceptable if the appliance does not overturn as a result of this test.		N
38	**Leakage Current Test**		**P**
38.1	The leakage current of a cord-connected portable, stationary, or fixed appliance, when tested in accordance with 38.3 – 38.7, shall be no more than:		P
	a) 0.5 milliampere for an ungrounded (two wire) portable, stationary, or fixed appliance;		P
	b) 0.5 milliampere for a grounded (three wire) portable appliance; and		N
	c) 0.75 milliampere for a grounded (three wire) stationary or fixed appliance.		N
38.2	Leakage current refers to all currents, including capacitively-coupled currents that may be conveyed		P

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Clause	Requirement + Test	Result - Remark	Verdict
	between exposed conductive surfaces of an appliance and ground or other exposed conductive surfaces of an appliance.		
38.3	All exposed conductive surfaces are to be tested for leakage current.		P
38.4	If a conductive surface other than metal is used for the enclosure or part of the enclosure, the leakage current is to be measured using a metal foil having an area of 10 by 20 centimeters in contact with the surface. Where the surface is less than 10 by 20 centimeters, the metal foil is to be the same size as the surface. The metal foil is not to remain in place long enough to affect the temperature of the appliance.		P
38.5	The measurement circuit for leakage current is to be as shown in Figure 38.1. The measurement instrument is described in (a) – (d). The meter used for a measurement need only indicate the same numerical value for a particular measurement as would the defined instrument. The meter used need not have all he attributes of the defined instrument.		P
38.6	A sample of the appliance is to be tested for leakage current starting with the as-received condition, but with its grounding conductor, if any, open at the attachment plug (open at receptacle as shown in Figure 38.1). The as-received condition is without prior energization, other than that which may have occurred as part of the production-line testing. The supply voltage is to be adjusted to 120 or 240 volts depending on the rating. Thermostats are to be closed.		P
38.7	Normally a sample will be carried through the complete leakage current test program as describe in 38.6, without interruption for other tests. With the concurrence of those concerned, the leakage current tests may be interrupted for the purpose of conducting other nondestructive tests.		P

39	**Leakage Current Test Following Humidity Conditioning**		**P**
39.1	A cord-connected appliance shall comply with the requirements for leakage current specified in 38.1 following exposure for 48 hours to air having a relative humidity of 88 ±2 percent at a temperature of 32 ±2°C (89.6±3.6°F).		P
39.2	To determine compliance with the requirement specified in 39.1, a sample of the appliance that has been preheated to a temperature just above 34°C (93.2°F) is to be contained in a chamber under the time, humidity, and temperature conditions specified. Following the conditioning, while still in the chamber, the sample is to be tested unenergized as described in 38.6(a). The sample, either in or immediately after removal from the chamber, is to be energized and tested as described in 38.6(b) and (c). The test is to be discontinued when the leakage current stabilizes or decreases.		P

40	**Immersion Protection Trip Time Measurement Test**		**N**
40.1	As-received hair dryers		N

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40.2	Conditioned hair dryers		N
41	**Dew Point Humidity Test**		**N**
41.1	Three samples of a hair dryer provided with an IDCI are to be conditioned in a chamber at a temperature of 5 ±2°C (41±3.6°F) for at least 4 hours and then transferred to a humidity chamber having a relative humidity of 86±2 percent at a temperature of 32±2°C (89.6±3.6°F). The transfer time is not to exceed 1 minute. The samples are to be energized by the insertion of their attachment plugs into receptacles of the voltage specified in 44.1.13. The on-off switch of the hair dryer is to be in the off position. The samples are to remain in the humidity chamber for 15 minutes. The results are acceptable if the IDCIs do not trip while in the chamber.		N
42	**Conductive Coating Test**		**N**
42.1	General		N
42.2	Thermal cycling		N
42.4	Short term aging		N
42.5	Humidity conditioning		N
42	Conductive Coating Test		N
42.1	General		N
42.2	Thermal cycling		N
42.4	Short term aging		N
42.5	Humidity conditioning		N
43	**Power Input Test**		**P**
43.1	The power input to an appliance marked with a rating of 50 watts or less shall be within the inclusive range of 75 – 110 percent of that rating. If the marked rating is greater than 50 watts, the power input shall be within the inclusive range of 90 – 110 percent of that rating.		P
43.2	With respect to 43.1, the wattage of an appliance marked with its electrical rating only in amperes and volts will be assumed to be the product of those two values.		P
43.3	The power input to the appliance is to be measured with the appliance at operating temperature under full-load conditions, and while connected to a circuit of a voltage in accordance with 44.1.13. Control switches or the equivalent, if provided, are to be set to give the maximum power input. For an appliance having a preheat cycle of operation as defined in 5.33, the maximum input value measured during the preheat cycle, with the appliance at room temperature at the beginning of the measurement, is to be used to determine compliance with the requirement specified in 43.1.		P
44	**Normal Temperature Test**		**P**
44.1	All appliances		P

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Clause	Requirement + Test	Result - Remark	Verdict
44.1.1	An appliance tested under the conditions described in this test shall not attain temperature rises at any time during the test greater than those indicated in Table 44.1.	No greater than those indicated in Table 44.1.	P
44.1.2	A temperature control that under intended operating conditions is relied upon to maintain temperatures within the limits specified in Tables 44.1 and 44.2, shall comply with the requirements in 54.2.1 for a combination temperature-limiting and temperature-regulating control.		P
44.1.3	At coils, the preferred method of measuring temperatures is the thermocouple method; temperature measurements by either the thermocouple or change-of-resistance method may be used. When temperatures of a coil or winding are measured by means of thermocouples, they are to be mounted on the outside of the coil wrap. If the coil is inaccessible for mounting thermocouples (for example, a coil immersed in sealing compound) or if the coil wrap includes thermal insulation such as more than 1/32 inch (0.8 mm) of cotton, paper, rayon, or similar insulation, the change of resistance method is to be used. For the thermocouple-measured temperature of a coil of an alternating-current motor (other than a universal motor) having a frame diameter of 7 inches (178 mm) or less (Table 44.1, item A, subitems 2 and 4), the thermocouple is to be mounted on the integrally applied insulation of the conductor.		N
44.1.4	In using the resistance method, the windings are to be at room temperature at the start of the test.		N
44.1.5	At a point on the surface of a coil where the temperature is affected by an external source of heat, the temperature rise measured by means of a thermocouple may be higher than the maximum indicated in Table 44.1 by the following amount		N
44.1.6	With respect to 44.1.5, if the coil wrap is not caused to exceed its temperature limitation by radiation from an external source, the temperature of the coil may be measured by means of a thermocouple on the integral insulation of the coil conductors.		N
44.1.7	All values for temperature rises in Tables 44.1 and 44.2 are based on an assumed ambient temperature of 25°C (77°F); however, tests may be conducted at an ambient temperature within the range of 20 – 30°C (68 – 86°F).		P
44.1.8	If the retention of the insulation of a heater cord depends upon a fabric braid, the braid shall not be removed nor subjected to a temperature rise of more than 65°C (117°F) unless other means are provided to hold the insulation in place. The jacket of Type HSJ or HSJO cord shall not be subjected to a temperature rise of more than 35°C (63°F) if the protection afforded by the jacket is required.		N
44.1.9	Certain special treatments, such as the use of an impregnant, have been determined to be acceptable for retaining the insulation around the conductors of a heater cord at elevated temperatures.		N
44.1.10	Thermocouples used to measure temperatures obtained by the thermocouple method are to consist of wires not larger than 24 AWG (0.221 mm2). The temperature is considered		P

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	to be stabilized when three successive readings, taken at intervals of 10 percent of the previously elapsed duration of the test but not less than 5 minutes indicate no change.		
44.1.11	When thermocouples are used in the determination of temperatures in connection with heating of electrical equipment, it is common practice to use thermocouples consisting of 30 AWG (0.05 mm2) iron and constantan wires and a temperature-indicating instrument. Thermocouples consisting of 30 AWG iron and constantan wires and a potentiometer-type temperature-indicating instrument are to be used whenever referee temperature measurements by thermocouples are required.		N
44.1.12	To determine whether an appliance complies with the requirement in 44.1.1, it is to be operated as follows. If the voltage rating of the appliance is within the range of 110 – 120 volts (inclusive), the test voltage is to be 120 volts. If the voltage rating of the appliance is within the range of 220 – 240 volts (inclusive), the test voltage is to be 240 volts. For an appliance having a voltage rating other than those previously specified, the test voltage is to be the marked voltage rating. Unless a particular voltage or other test condition is specified, the test voltage is to be increased, if necessary, to cause the wattage input to the appliance to be equal to its marked wattage rating.	Test at 125V	P
44.1.13	If an appliance uses a motor in addition to a heating element, the voltage applied to an integrally connected motor is to be 120 volts for an appliance rated at 110 – 120 volts, 240 volts for an appliance rated at 220 – 240 volts, or the rated voltage of the appliance for other cases. A motor supplied from a separate circuit is to be operated at a voltage (depending upon the motor rating) as specified for an integrally connected motor.		N
44.1.14	During the test, each general use receptacle, or a general use receptacle intended for a limited current load, shall be loaded with a 15-ampere resistive load or with a lesser load if marked in accordance with 72.1.6.		N
44.1.15	The appliance is to be mounted or supported as in service and tested under conditions approximating those of intended operation. If a timer switch or the equivalent is provided as part of the appliance, an appropriate cycle of operation shall be used.		P
44.1.16	A manually resettable thermal device or a thermal cutoff shall not operate during the normal temperature test.		N
44.1.17	In a hand-supported hair dryer, the motor circuit shall not become de-energized during the normal temperature test.		N
44.1.18	A means for adjusting the operating temperature is to be set to give maximum heating.		N
44.1.19	An electrical heating element intended for application to the hair is to be loaded with a moistened cloth and then operated until the moisture has been evaporated and the heating surface of the unit has attained a temperature of 204°C (400°F). Following a 2 minute period with the unit disconnected, during which it is to be reloaded with another moistened cloth, the heating and evaporating operation is to be conducted a second time. The complete cycle is then		N

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Clause	Requirement + Test	Result - Remark	Verdict
	to be repeated again. Temperatures are to be measured throughout the test		
44.1.20	An appliance having a reservoir for heating water shall have the water temperature measured by means of a thermocouple floated approximately 3/16 inch (4.8 mm) beneath the surface of the solution and located midway between the outer surface of the electrode enclosure and the inner surface of the water reservoir. The unit is to be tested in a room ambient of 25°C (77°F).		N
44.1.21	For an appliance in which clips to be applied to the hair are heated by an external heater, the test is to consist of operation of the appliance until temperatures are constant, with the clips in place on the heater.		P
44.1.22	With regard to the Exception to 44.1.1, the equivalent continuous normal use temperature is to be determined as follows.		P
44.2	Wax depilatory appliances		N
44.2.1	The appliance is to be loaded with the maximum recommended amount of wax and operated continuously until constant temperatures have been reached. An adjustable temperature control is to be set for maximum heating. If the appliance has several heat settings for different functions (as noted in 9.3.3), it is to be operated at the highest heat setting, as well as at the maximum setting intended to maintain the molten wax at the temperature for application to the skin.		N
44.2.2	The wax temperature is to be measured by means of a thermocouple immersed beneath the surface of the wax to a depth of approximately one-half of the total depth, at the approximate center of the reservoir. The wax is to be slowly and continuously stirred while temperatures are being recorded. For depilatory appliances having self-contained wax applicators (no open reservoirs), thermocouples are to be inserted into the wax applicators to a depth of approximately one-half of the total depth of the wax.		N
44.2.3	In addition to complying with the requirements specified in 9.3.1, 9.3.2, and 44.1.1, the visible overheat condition indicator, when required as specified in 9.3.3(b), shall function when the wax temperature exceeds 75°C (167°F).		N
44.3	Heated air curling irons and brushes		N
44.3.1	A heated air curling iron or brush is to be operated continuously with all air intake and outlet openings unrestricted until temperatures stabilize. The appliance is then to be operated through 30 cycles, with one cycle consisting of one minute of operation with the air intake and outlet openings blocked as described in 44.3.2, followed by 10 seconds of operation with all air openings open.		N
44.3.2	With regard to 44.3.1, air intake openings provided in the gripping area of the handle that may be blocked by the user’s hand, such as those shown in Figure 44.2, are to be blocked such that all openings in three or fewer quadrants		N

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Clause	Requirement + Test	Result - Remark	Verdict
	of the handle circumference will be blocked. (If there are openings in all quadrants, those in one quadrant are to be left open. Openings provided in the base of a cylindrica lhandle, such as those shown in Figure 44.3, are not considered likely to be blocked by the user's hand and are not to be blocked.) Three-fourths of the air outlet openings in the barrel are also to be blocked. For example, in a unit provided with eight parallel rows of openings in the length of the barrel, two adjacent rows are to be left unblocked.		
44.4	Hair-drying appliances		N
44.4.1	For a floor- or table-supported hair dryer having a bonnet or rigid hood and provided with an adjustable temperature control, temperatures are to be recorded after 15 minutes of operation with the control set for maximum heating and with the dummy head in the position normally occupied by the human head under the dryer.		N
44.4.2	For a hair dryer that is provided with a rigid hood, the dummy head is to be located so that its surface will be a minimum of 1 inch (25.4 mm) from the interior surface of the hood and so positioned in the hood that maximum temperatures will result.		N
44.4.3	A floor-supported hair dryer not provided with an adjustable temperature control is to be operated continuously (with the dummy head in place) until all temperatures become constant.		N
44.5	Hand-supported hair dryers		N
44.5.1	A hand-supported hair dryer shall be tested with the adjustable temperature control, if any, set for the most severe condition of use and supported in the position representing the most severe conditions of use. The dryer shall be operated continuously until stabilized conditions are achieved, first without any attachment on the heated air outlet nozzle, and then, if the dryer is provided with one or more attachments for the heated air outlet, in turn with each attachment in place, as intended.		N
44.5.2	A dryer with screen-covered air inlet openings shall have two layers of fabric loosely secured over the openings to simulate the accumulation of hair, lint, or other particulate matter. The fabric shall be white, 100 percent untreated cotton terrycloth having a pile weave and a nominal weight of 8 ounces per square yard (271 g/m2).		N
44.5.3	If an automatically-resetting thermostat operates during the test procedure described in 44.5.2, the test is to continue for no less than 5 cycles, or 3 hours (whichever is less) to determine maximum temperatures under those conditions. The test is then to be repeated with only one layer of fabric over the air inlet openings. If the thermostat operates under this condition, the test is to continue for no less than 3 cycles, or 1 hour (whichever is less) to determine maximum temperatures under those conditions. The test is then to be repeated with no fabric. Operation of the thermostat under this condition is not		N

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Clause	Requirement + Test	Result - Remark	Verdict
	permitted.		
44.5.4	During each of these tests, the plane of the thermocouple grid specified in 44.5.5 is to be positioned 1 inch (25.4 mm) from the plane of the heated air outlet of:		N
	a) The dryer nozzle or		N
	b) The attachment nozzle.		N
44.5.5	The thermocouple grid assembly is to consist of two pieces of 1/16-inch (1.6-mm) thick glass epoxy board of the configuration and dimensions shown in Figure 44.4. The two boards are to be separated 1/8-inch (3.2-mm) by one 5-1/4- by 1/4- by 1/8-inch (133- by 6.4- by 3.2-mm) wood spacer at the top and bottom edges. Each spacer is to be secured by four4–40by3/8-inch countersunk flat head machine screws.		N

45	**Dielectric Voltage-Withstand Test**		**P**
45.1	An appliance, while heated to its operating temperature, shall be subjected for 1 minute to a 60 hertz essentially sinusoidal potential as specified in Table 45.1 applied between live parts and exposed dead-metal parts. There shall not be dielectric breakdown.		P
45.2	With respect to 45.1, an appliance that has no exposed dead-metal parts is to be closely wrapped in metallic foil and the test potential is to be applied between the foil and all live parts.		P
45.3	To determine compliance with the requirements in 45.1 and 45.2, the test is to be made using a 500 volt-ampere or larger capacity testing transformer, the output voltage of which is essentially sinusoidal and can be regulated. The applied potential is to be increased from zero until the required value is reached and is to be held at that value for 1 minute. The increase in the applied potential is to be at a substantially uniform rate and sufficiently rapid to be consistent with its value being correctly indicated by a voltmeter.		P
45.4	An appliance provided with a normally-open immersion protective device is to be tested as follows. The appliance is to be connected to a supply circuit of rated voltage, the protective device is to be reset (closed), an on-off switch of the appliance is to be in the ″off″ position, and the test potential is to be applied between each supply circuit conductor and exposed dead-metal parts as described in 45.1 – 45.3.		N

46	**Abnormal Operation Tests**		**P**
46.1	General		P
46.1.1	A hair dryer shall not cause ignition of any material or emission of flame, sparks, molten metal, or similar result		N
46.1.2	In a test to determine compliance with the requirement in 46.1.1, the dryer is to be connected to a supply circuit of a voltage in accordance with 44.1.13. Temperature-control adjustments or the equivalent, if any, are to be set in the position that will result in the most		N

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Clause	Requirement + Test	Result - Remark	Verdict
	severe test, and thermostats or other temperature controls are to be shunted out of the circuit unless it has been determined that they are rugged, reliable, and not likely to be defeated by the user.		
46.1.3	The test procedures described in 46.2.1 – 46.4.5.1 are for a conventional-type appliance.		P
46.1.4	A curling iron is to be operated without any separable stand on a softwood surface covered with two layers of white tissue paper. The curling iron is to be covered loosely with a double layer of cheesecloth as described in 46.2.1.		N
46.1.5	A steam-type curling iron is to be energized at rated wattage and the water reservoir filled with a solution of hard water (1/2 gram of calcium sulfate, CaSO42H20, per liter of distilled water) in an amount equal to the capacity of the reservoir.		N
46.1.6	A steam-type curling iron or mist-type hair styler having an integral liquid reservoir shall be subjected to an overfill test consisting of pouring a solution of hard water as described in 46.1.5 into the reservoir.		N
46.1.7	In order to determine if a steam-type hair curler having an integral liquid reservoir will need to have the wording specified in 74.7(g)(16), the appliance shall be subjected to a water droplet test consisting of pouring a solution of hard water, as described in 46.1.5, into the reservoir.		N
46.2	Hair dryers – drape test		N
46.2.1	A bonnet-style hair dryer shall be operated without the use of a dummy head until all temperatures stabilize, and then draped with a double layer of cheesecloth.		N
46.2.2	A handheld hair dryer shall be placed on a flat, horizontal softwood surface, operated until all temperatures stabilize, and then draped with a double layer of cheesecloth.		N
46.2.3	The cheesecloth shall be draped in a manner so as to retard the air flow and cover the hottest area of the hair dryer but not be deliberately manipulated to cause an overly restrictive air flow		N
46.2.4	The hair dryer shall be operated for 8 hours or until stabilized conditions are apparent, whichever is longer. The cheesecloth shall not discolor, glow, or flame as a result of this test.		N
46.3	Hair dryer locked rotor test		N
46.3.1	The sample from the drape test shall be used for this test.		N
46.3.2	The motor blower shall be locked and a single layer of cheesecloth loosely draped over the appliance as described in 46.2.3. The dryer is then to be operated for 7.5 hours or until stabilized conditions are apparent, whichever is longer.		N
46.3.3	A second handheld hair dryer sample shall be subjected to the conditions in 46.3.2 while mounted with the air outlet nozzle pointed downward 45 degrees from vertical.		N
46.3.4	The tests in 46.3.2 and 46.3.3 shall be conducted at each heat setting. A new sample shall be used if a limit control operates to render a sample inoperative prior to		N

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Clause	Requirement + Test	Result - Remark	Verdict
	completing all tests		
46.4	Hand-supported hair dryers		P
46.4.1	Softwood surface temperature test		P
46.4.2	Restricted air inlet test		P
46.4.3	Restricted air outlet test		P
46.4.4	Floor drop test		P
46.4.5	Broken heating element test		P
46.4.6	Motor slowdown test		P
46.5	Dual-voltage appliances		N
46.5.1	In addition to the applicable tests described in 46.2.1 – 46.4.5.1, a dual-voltage appliance shall be subjected to the tests described in 46.5.2 – 46.5.4. These tests are subject to the test conditions described in 46.1.2 and the acceptance criteria described in 46.1.1. There shall be no electrical or mechanical breakdown of the voltage selector switch.		N
46.5.2	The appliance shall have its voltage selector set in any marked supply-circuit voltage position with the equipment connected to any one of the rated supply circuits. The combination of selector settings and supply circuit to which the equipment is connected is to be that which develops the most severe operating conditions.		N
46.5.3	If provided, an externally operable input voltage selector is to be operated for 25 cycles with the appliance operating at the minimum rated voltage and for 25 cycles with the appliance at the maximum rated voltage. Each cycle is to consist of moving the voltage selector to its alternate position and back at a rate of 6 cycles per minute, with the voltage selector in each position for 5 seconds. The operating and temperature controls are to be set so as to result in the most adverse operating conditions.		N
46.5.4	For an externally operable voltage selector switch that interlocks with the power switch and cannot be operated with the power switch in the on position, the voltage selector is to be operated for 25 cycles each at the maximum and the minimum rated voltages.		N
46.6	All appliances – short-circuit, stall tests		P
46.6.1	A motor in a limited-energy circuit is to be short-circuited and, as a separate test, is to be stalled. A motor in a low-voltage circuit is to be stalled. Any solid-state device, such as a rectifier, a transistor, a resistor, or a capacitor, is to be subjected to the tests described in 46.6.2 and 46.6.3.		P
46.6.2	If an appliance uses one or more solid-state devices such as a rectifier, a transistor, a resistor, or a similar component, no condition that involves a risk of fire, electric shock, or injury to persons shall develop when the circuit between any two terminals of any such component is opened or shorted. If the appliance uses a capacitor in combination with one of the above-specified components, no condition that involves a risk of fire, electric shock, or injury to	No any risks.	P

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Clause	Requirement + Test	Result - Remark	Verdict

	persons shall develop when the capacitor is short-circuited. Only one of the simulated fault conditions described is to be imposed at one time.		
46.6.3	Short-circuit tests to determine compliance with the requirements in 46.6.2 are to take into account the intended usage of the appliance.		P
46.6.4	If an appliance is provided with means for controlling speed, the test is to be conducted at both the maximum and minimum speed settings of the control, and may be conducted at interim speed settings.		N
46.7	Hair curler heater – short-circuit test		P
46.7.1	If a hair curler heater uses one or more automatic resetting thermostats, the thermostats are to be short-circuited and the appliance is to be operated under the conditions described in 46.7.2. Six samples of the appliance shall be tested.		P
46.7.2	Three samples of the hair curler heater are to be tested with the curlers in place, and three samples are to be tested without the curlers.		P
46.7.3	The results of the test described in 46.7.2 are acceptabl		P
46.7.4	A manually-resettable protector or a replaceable thermal cutoff (fusible link) used to provide compliance with the requirement in 46.7.1 shall not function during the normal temperature test.		P
46.8	Bonnet-type hair dryers – hair entanglement tes		N
46.8.1	A bonnet- or helmet-type hair dryer with heater and blower integral with the head piece shall be subjected to the test described in 46.8.2.		N
46.8.2	A sample of the hair dryer is to be mounted on a stand and connected to a 120-volt, 60-hertz source of power supply.		N
46.9	Wax depilatory appliances		N
46.9.1	If a wax depilatory appliance uses one or more automatic reset temperature controls, all such controls are to be short-circuited and the appliance is to be operated under the conditions described in 46.9.2 and 46.9.3.		N
46.9.2	The appliance is to be operated empty and also with the maximum recommended amount of wax. A movable part or cover is to be in the intended position resulting in the most adverse conditions. A self-closing cover (as described in 9.3.4) is to remain in its closed position.		N
46.9.3	One sample is to be tested under each condition in 46.9.2. Each sample is to be placed on a white, tissue-paper-covered soft pine wood surface in a draft-free location.		N
	tissue-paper-covered soft pine wood surface in a draft-free location.		N
46.9.4	The results are acceptable		N
46.9.5	An abnormal test is also to be conducted by operating the appliance under the conditions of intended use, as described in 44.2.1 and 44.2.2, but defeating the temperature control that operates to keep the wax temperature at or below 75°C (167°F). The visible overheat condition indicator specified in 9.3.5 shall function when the wax temperature exceeds 75°C.		N

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Clause	Requirement + Test	Result - Remark	Verdict
46.10	Hair dryer immersion protective devices with convenience receptacles		N
46.10.1	To determine compliance with 7.9(h), the hair dryer immersion protective device shall not present a risk of fire or electric shock when tested as described in 46.10.2 – 46.10.5. Additionally, if the immersion protective device is functional at the end of the test, it shall comply with the applicable requirement in the high-resistance ground faults test specified in the Standard for Ground-Fault Circuit-Interrupters, UL 943.		N
46.10.2	With regard to the requirement specified in 46.10.1, a risk of fire or electric shock is considered to exist		N
46.10.3	The immersion protective device is to be plugged into a duplex receptacle. The outlet's face of the duplex receptacle is to be in a vertical plane.		N
46.10.4	If the 20-ampere line fuse, appliance protective device, or any other circuit component opens as indicated in 46.10.3(b), (c), or (d), the load on the convenience receptacle is to be adjusted to the highest value that will permit the test to be conducted for 7 hours or a Minimum of 1 hour with stabilized conditions.		N
46.10.5	If operation of the supervisory circuit indicates that the immersion protective device is functional,the device shall comply with the applicable requirement in the high-resistance ground faults test specified in the Standard for Ground-Fault Circuit-Interrupters, UL 943.		N
46.11	Appliances having an automatically-controlled preheat cycle		N
46.11.1	Six samples of a hair curler heater (hair setter) and three samples of other appliances are to be tested. Three samples of a hair curler heater are to be tested with the curlers in place and three samples are to be tested without the curlers.		N
46.11.2	The results are acceptable		N

47	**Exposure to Moisture Test**		**N**
47.1	An appliance that may alternately be used wet and dry (such as a hair dryer-styler having comb and brush accessories that may be used for setting or styling of wet or damp hair and then used as a dryer, or a hair untangler) shall be tested as described in 47.2 – 47.6.		N
47.2	One sample is to be subjected to this test in an unenergized condition. Any attachment is to be oriented to result in the most unfavorable condition of use.		N
47.3	Each sample is to be oriented above a standard test solution (consisting of 1/2 gram of calcium sulfate per liter of water) so that the comb teeth or the center row (or rows) of brush bristles are pointing vertically downward.		N
47.4	Following the dipping cycles, and while being held in its final vertical position, the sample is to be completely and closely wrapped in metal foil.		N
47.5	The results are acceptable if the appliance, in an unenergized condition, withstands for 1 minute without breakdown a 60-hertz essentially sinusoidal potential of 2,500 volts applied between live parts and		N

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Clause	Requirement + Test	Result - Remark	Verdict

	the foil wrapping.		
47.6	If the appliance is supplied with one or more accessory attachments, the complete moisture test is to be conducted using each accessory. Samples of the appliance are to be in operating condition for this test.		N

48	**Strain Relief Test**		**P**
48.1	The strain relief means provided on an attached flexible cord, when tested in accordance with 48.2, shall withstand for 1 minute a pull of 35 pounds force (156 N) applied to the cord.		P
48.2	The connections of the cord inside the appliance are to be disconnected. The specified force is to be applied to the cord and so supported by the appliance that the strain relief means will be stressed from any angle that the construction of the appliance permits. At the point of disconnection of the conductors, there shall not be movement of the cord to indicate that stress on the connections would have resulted.		P

49	**Cord Flexing Test**		**P**
49.1	Each of six″as received″samples of a hand-supported hair-drying appliance (such as a hair dryer, blower-styler, heated air comb or brush, hair dryer-curling iron combination, wall-hung hair dryer, or the hand unit of a wall-mounted hair dryer), comb, curling iron, untangler, hair crimping iron, hair straightening iron, or similar hand-supported appliance shall be subjected to a cord flexing test as described in 49.2.		P
49.2	Each sample is to be mounted in a guide with a 1/4 pound (113 g) weight attached to the cord 8 inches (203 mm) from the cord entry hole so that the unit can be rotated 540 degrees about the axial center of the cord.		P
49.3	For an appliance using a cord swivel construction, the test described in 49.2 is to be conducted with the swivel locked in place.		P
49.4	For both″as received″and conditioned samples, test results are acceptable		P
49.5	An appliance using a cord swivel construction and tested with the swivel operating is to be tested with and without the weight, using separate sets of six samples for each condition, with the cord hanging freely during the test.		P
49.6	Three samples of a hair-drying appliance are to be conditioned in an air oven maintained at a temperature of 100°C (212°F) for 96 hours or at 87°C (189°F) for 168 hours, as specified by the manufacturer. Following oven conditioning and cooling to a room temperature of 23±2°C (73±3.6°F), the samples are to be tested as described in 49.2.		P

50	**Cord Flexing Test for Appliance Leakage-Current-Interrupter (ALCI)**		**P**
50.1	An appliance leakage-current-interrupter (ALCI) configured as an attachment plug cap shall be subjected to the cord flexing test described in 50.2 –		P

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Clause	Requirement + Test	Result - Remark	Verdict
	50.4. There shall be no damage to the power supply cord insulation or appliance leakage-current-interrupter enclosure and no loss of continuity in any power supply conductor.		
50.2	To conduct this test, the specified units of the ALCI are to be assembled to the test fixture shown in Figure 50.1 so there is no interference with the test procedure. Each unit is to be mounted with the point of cord entry into the ALCI at the center of rotation.		N
50.3	Six units are to be subjected to 6,000 cycles of flexing at a rate of 10 cycles per minute in the plane of the face of the ALCI. The test is to be repeated using six additional units with the flexing in the direction perpendicular to the plane of the face of the ALCI.		N
50.4	With reference to 50.3, each cycle consists of a 90-degree rotation of the unit in one direction; a 180-degree rotation in the opposite direction; and then a return to the starting point.		N

51	**Test for Security of Swivel Assembly**		**P**
51.1	The supply swivel assembly on one as-received sample and three conditioned samples (as described in 51.2) are each to be subjected to a direct pull force of 35 pounds (156 N) for 1 minute with the force applied at any angle the construction of the appliance will permit.		P
51.2	With regard to 51.1, the conditioned samples are to be maintained for 7 hours at a temperature of 10°C (18°F) higher than the temperature measured on the swivel assembly during the normal temperature test, but not less than 70°C (158°F)		P

52	**Swivel Endurance Test**		P
52.1	An appliance provided with a cord swivel shall be subjected to the cord swivel endurance test described in 52.2 – 52.6.		P
52.2	If a hand-supported comb, curling iron, hair untangler, or similar hand-supported appliance is provided with a cord swivel, each of the same six samples subjected to the Cord Flexing Testing, Section 49, under the condition of the swivel operating are to be subjected to the tests described in 52.3 and 52.5. For a hand-supported hair dryer provided with a cord swivel, six new samples are to be tested as described in the exception of 52.3 and in 52.5.		P
52.3	The cord flexing test described in 49.2 is to be continued and the swivel is to be cycled for the additional number of cycles required to total 100,000 cycles. (A cycle consists of 540 degrees in one direction plus 540 degrees in the reverse direction back to the starting point).		P
52.4	A hand-supported hair dryer that can be converted into a curling iron (for example, by use of a hair-curling attachment) is to be cycled in accordance with 52.3, with the appliance in the curling iron configuration and, in accordance with the Exception to 52.3, with the appliance in the hair dryer configuration. A separate set of six samples is to be used for testing in each configuration.		P

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Clause	Requirement + Test	Result - Remark	Verdict
52.4	A hand-supported hair dryer that can be converted into a curling iron (for example, by use of a hair-curling attachment) is to be cycled in accordance with 52.3, with the appliance in the curling iron configuration and, in accordance with the Exception to 52.3, with the appliance in the hair dryer configuration. A separate set of six samples is to be used for testing in each configuration.		N
52.6	Test results are acceptable		P
53	**Hinge Endurance Test**		**N**
53.1	A hair crimping iron, a hair straightening iron, or an appliance such as a hair dryer having a foldable handle shall be subjected to the hinge endurance test described in 53.2 – 53.4.		N
53.2	Three samples are to be energized at the voltage specified in 44.1.13. Each sample of a hair crimping or hair straightening iron is then to be subjected to 30,000 cycles of opening and closing the appliance.		N
53.3	At the conclusion of the cycling described in 53.2, a dielectric voltage-withstand test as described in the Dielectric Voltage-Withstand Test, Section 45, is to be conducted. The test potential is to be applied between live parts and exposed surfaces of the hinge assembly.		N
53.4	The results of the cycling described in 53.2 are acceptable if, upon completion of the required number of cycles, the samples are operable		N
54	**Test of Automatic Controls**		**N**
54.1	Overload		N
54.2	Endurance		N
55	**Test of Thermal Cutoffs (Fusible Links)**		**N**
55.1	A thermal cutoff shall open the circuit in the intended manner without causing the short circuiting of live parts and without causing live parts to become grounded to the enclosure		N
55.2	In the case of a hand-supported hair dryer using a thermal cutoff, each of five samples shall be tested with the dryer oriented in the position		N
55.3	To determine whether a thermal cutoff complies with the requirement in 55.1, the appliance is to be operated five times as indicated, and it is required that the cutoff perform acceptably each time.		N
55.4	The opening temperature of a thermal cutoff shall not differ by more than 8.3°C (15.0°F) from the rated opening temperature. A thermal cutoff shall be investigated with respect to its aging characteristics and its ability to open without a risk of fire, electric shock, or injury to persons under overload and short-circuit conditions.		N
56	**Motor Control Overload Test**		**P**

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Clause	Requirement + Test	Result - Remark	Verdict
56.1	A motor control device supplied as a part of an appliance and not having a horsepower rating equivalent to the motor it controls, shall be capable of performing effectively when subjected to an overload test consisting of 50 cycles of operation, making and breaking the stalled rotor current of the motor. Dielectric or Mechanical breakdown of the device or undue pitting or burning of the contacts shall not occur.		P
56.2	To determine whether a motor control device complies with the requirement in 56.1, the device is to be tested with the appliance connected to a supply circuit of rated frequency and a voltage in accordance with 44.1.13, and with the rotor of the motor locked in position.		P
57	**Grounding Continuity Test**		**N**
57.1	The resistance of the grounding path between a dead-metal part of an appliance as specified in 27.3 and the equipment grounding terminal or lead or the point of attachment of the wiring system or the grounding blade of an attachment plug shall be no more than 0.1 ohm.		N
57.2	With reference to 57.1, the resistance may be determined by any convenient method. If the results do not comply with the requirement specified in 57.1, either a direct- or alternating-current at a potential of no more than 12 volts, and equal to the current rating of the maximum-current-rated branch circuit overcurrent protective device that may be used with the appliance, is to be passed from a dead-metal part to either the:		N
	a) Equipment grounding terminal,		N
	b) Point of attachment of the wiring system, or		N
	c) Grounding blade of the attachment plug.		N
58	**Test for Permanence of Cord Tag for Hand-Supported Hair-Drying Appliances**	No Hair-Drying Appliances.	**N**
58.1	General		N
58.2	Test conditions		N
58.3	Test method		N
59	**Mounting Means Strength Test**		**N**
59.1	To determine compliance with 12.1 after the appliance is installed, the appliance is to be mounted in accordance with the manufacturer's installation instructions using fasteners and constructions as described. If no wall constructions are specified, 3/8-inch (9.5-mm) thick plasterboard – drywall on 2 by 4 studs at 16-inch (406.4-mm) centers – is to be used as the support surface.		N
60	**Extended Operation Test**		**N**
60.1	To determine if a higher temperature rise than that specified in Table 44.1 is acceptable for a fiberglass sleeving (see Note (h) of Table 44.1), three		N

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Clause	Requirement + Test	Result - Remark	Verdict
	samples of the appliance are to be tested as described in 60.2 – 60.5.		
60.2	Each sample is to be operated continuously for 1,000 hours. The test voltage (as specified in 44.1.13) is to be increased, if necessary, to cause the wattage input to the appliance to be equal to its marked wattage rating.		N
60.3	Each sample, while at room temperature, is then to be subjected to impacts as described in 60.4 – 60.6. After each drop for a hand-supported appliance and the one impact for other types of appliances, the leakage current test is to be repeated. After the leakage current test, each sample is to be subjected to the Dielectric Voltage-Withstand Test, Section 45. For a hand-supported appliance, the dielectric voltage-withstand test is to be conducted after the final leakage current test.		N
60.4	Each of three samples of a hand-supported portable appliance is to be dropped 3 feet (0.91 m) to strike a hardwood surface in the position most likely to produce adverse results.		N
60.5	Stationary, fixed, counter-supported, or floor-supported appliances are to be subjected to the ball impact test described in 60.6.		N
60.6	Each of three samples of the appliance is to be subjected to a single impact of the value specified in Table 60.1 for the applicable appliance type, on any surface that can be exposed to a blow during normal use.		N
60.7	The results are acceptable		N

61	**Heating Element Endurance Test**		**P**
61.1	Each of six samples of an appliance having an automatically controlled preheat cycle are to be subjected to 6,000 cycles of operation. Each cycle is to consist of energizing the appliance at the test voltage specified in 44.1.13 from room temperature to the maximum stabilized temperature condition, then de-energizing and cooling to room temperature.		P

62	**Test of Physical Properties of a Liquid Container, Seal, or Diaphragm**		**N**
62.1	If physical deterioration of a liquid container, seal, diaphragm, or similar part would result in a risk of fire or electric shock, the component shall be tested to determine its resistance to deterioration from the liquid intended to contact it.		N
62.2	The test procedure for determining whether a component complies with the requirements specified in 62.1 depends upon the material of which it is composed, its size and shape, the mode of application in the appliance, and similar criteria.		N
62. 3	With respect to 62.1 and 62.2, a component of rubber, neoprene, or thermoplastic material shall be tested to compare its tensile strength and elongation before and after artificial aging.		N
62.4	As an alternative to the air oven aging specified in Table 62.1, the acceptability of a liquid container, seal, or diaphragm may be determined by means of an aging		N

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Clause	Requirement + Test	Result - Remark	Verdict
	test on the complete appliance under service conditions.		
63	**Label Adhesion Test**		**P**
63.1	To determine if a pressure-sensitive label or a label secured by cement or adhesive meets the requirements for its intended use, representative samples that have been subjected to the tests specified in 63.2 – 63.7 shall meet all of the following conditions:		P
	a) Each label shall demonstrate strong adhesion and the edges shall not be curled,		P
	b) The label shall resist defacement or removal as demonstrated by scraping across the test panel with a flat metal blade 1/32 inch (0.8 mm) thick held at a right angle to the test panel, and		P
	c) The printing shall be legible and shall not be defaced by rubbing with thumb or finger pressure.		P
63.2	For each of the types of conditioning specified in 63.3 – 63.7, three samples of a label are to be applied to the same test surface used in the intended application. The labels are to be applied to the test surface no less than 24 hours prior to testing.		P
63.3	Three samples of the labels are to be investigated as received.		P
63.4	Investigation of samples at the end of each test as indicated in 63.5 – 63.7 is to be made:		P
	a) Immediately following removal from each test medium and		P
	b) After exposure to room temperature for 24 hours following removal from each test medium.		P
63.5	Three samples of the labels under test are to be placed in a full-draft circulating-air oven maintained at the temperature indicated in Table 63.1 for 240 hours.		P
63.6	Three samples of the labels under test are to be immersed in water at a temperature of 23.0±2.0°C (73.4±3.6°F) for 48 hours.		P
63.7	If the labels are exposed to unusual conditions in service, such as exposure to medicant, detergents, oil, or other substances, three additional samples are to be conditioned as follows. The samples are to be immersed in a solution representative of service use, maintained at 23.0 ±2.0°C (73.4±3.6°F) for 48 hours.		N
64	**Flammability Test – Wax for Depilatory Appliances**		**N**
64.1	The wax for a depilatory appliance shall comply with the flammability test described in 64.2.		N
64.2	The depilatory appliance is to be loaded with the maximum recommended amount of wax and operated at the voltage specified in 44.1.13 and the highest heat setting until the molten wax is at its maximum temperature. A wooden match is then to be lighted and touched against the surface of the wax. The wax in the reservoir shall not ignite.		N
65	**Additional Testing for Hand-Supported Grooming Appliances with a Detachable Power Supply Cord and for Hand-Supported Grooming Appliances with Detachable Parts Intended to be Disconnected From Power**		**N**

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Clause	Requirement + Test	Result - Remark	Verdict

	Under Load Conditions		
65.1	Hand-supported appliances provided with a detachable power supply cord and hand-supported appliances with detachable parts intended to be disconnected from power under load conditions shall additionally comply with 65.1 – 65.11.		N
65.2	Appliances with detachable power supply cords shall withstand the impact described in 65.3 with the power supply cord attached and detached from the appliance. A hand-supported appliance, when detached from its base or stand, shall withstand the drop impact test described in 65.3. Compliance shall be determined after impact(s) as follows:		N
	a) Uninsulated live parts shall not be accessible to contact as determined by the probes in Figures 8.1 and 13.3;		N
	b) An operating condition shall not arise that affects the mechanical performance of the appliance including compliance with (a) and (b) of the Exception to 13.3.1.2; or		N
	c) The electrical insulation shall comply with the dielectric voltage-withstand requirements of Section 45.		N
65.3	A hand-supported appliance or hand-supported part of an appliance shall be subjected to the drop impact test described in (a) and (b).		N
65.4	A sample of the appliance constructed as detailed in 13.3.1.14 shall be subjected to the testing detailed in 65.5		N
65.5	A sample of the appliance shall be connected to a supply adjusted to rated voltage and operated under conditions of maximum loading.		N
65.6	A sample of the appliance constructed as detailed in 13.3.1.15 shall be subjected to the testing detailed in 65.7		N
65.7	A sample of the appliance shall be connected to a supply adjusted to rated voltage and operated under conditions of maximum loading.		N
65.8	To determine compliance with 13.3.1.16, an appliance shall be placed on a supporting surface inclined at 10 degrees from the horizontal and is to be oriented to the position most likely to cause tipping.		N
65.9	After completion of the test of 65.8 and with the base of the appliance still at a 10 degree angle, the hand-supported part of the appliance shall be removed and inserted ten complete times.		N
65.10	A sample of the appliance provided with the locking mechanism required by 13.3.1.17 shall be subjected to the unenergized mechanical testing detailed in 65.11		N
65.11	The locking mechanism shall be subjected to 10,000 cycles of assembly/disassembly at a rate not to exceed 10 cycles per minute.		N

66	**Thermoplastic Motor Insulation Systems**		N
66.1	General		N
66.2	Abnormal conditioning		N
66.3	Overload-burnout conditioning		N

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Clause	Requirement + Test	Result - Remark	Verdict
66.4	Thermal aging		N
67	**Ozone Test**		**N**
67.1	Grooming appliances employing ionization circuitry shall not produce a concentration of ozone exceeding 0.05 parts per million by volume when tested as described in 67.2 – 67.7.		N
67.2	The test is to be conducted in a room having a volume of 950 – 1100 cubic feet (26.9 – 31.1 m3) with a minimum side dimension of 8 feet (2.4 m) and a maximum height dimension of 10 feet (3.0 m) without openings.		N
67.3	During the test, the test room is to be maintained at a temperature of 25±2°C (77±4°F) and a relative humidity of 50 ±5 percent.		N
67.4	The product is to be located in the center of the test room floor for floor supported product and about 30 – 36 inches (762 – 910 mm) above the floor for table-mounted or hand-held product.		N
67.5	The ozone monitor sampling tube is to be located 2 inches (50 mm) from the air outlet of the product and is to point directly into the air stream.		N
67.6	The product shall be operated at maximum ozone output and the emission of ozone shall be monitored for 7 hours to determine the concentration.		N
67.7	If the ionizing can be energized with the fan not functioning or with particle filters removed, the test described in 67.2 – 67.6 is to be repeated with the fan not operating or with particle filters removed.		N
MANUFACTURING AND PRODUCTION-LINE TESTS			
68	**Dielectric Voltage-Withstand Test**		**P**
68.1	As a routine production-line test, each appliance shall be subjected to the application of a potential at a frequency within the range of 40 – 70 hert		P
68.2	Each power supply cord terminating in a swivel assembly shall be subjected to a test potential between:		P
	Line conductors and Each line conductor and grounding conductor, if provided.		N
68.3	The production-line test described in 68.1 shall be in accordance with either condition A or B of Table 68.1. The results are acceptable if there is no dielectric breakdown.		P
68.4	The appliance may be in a heated or unheated condition for the test.		P
68.5	The test described in 68.1 shall be conducted when the appliance is complete (fully assembled). It is not intended that the appliance be unwired, modified, or disassembled for the test.		N
68.6	When the appliance uses a solid-state component that is not relied upon to reduce the risk of electric shock and that can be damaged by the dielectric potential, the test described in 68.1 may be conducted before the component is electrically connected, only when a random sampling of daily production is tested at the potential specified in Table 68.1. The circuitry may be rearranged for the purpose of		N

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Clause	Requirement + Test	Result - Remark	Verdict

	the test to reduce the likelihood of solid-state component damage while retaining representative dielectric stress of the circuit.		
68.7	The test equipment shall include:		N
	a) A transformer having an essentially sinusoidal output,		N
	b) A means of indicating the test potential		N
	c) An audible or visible indicator of dielectric breakdown, and		N
	d) Either a manual reset device to restore the equipment after dielectric breakdown or an automatic reject feature activated by any unit indicating dielectric breakdown.		N
68.8	If the output of the test equipment transformer is less than 500 volt-amperes, the equipment shall include a voltmeter in the output circuit to directly indicate the test potential.		N
68.9	If the output of the test equipment transformer is 500 volt-amperes or larger, the test potential shall be indicated		N
68.10	Test equipment other than that described in 68.7 – 68.9 may be used if the intended factory test is accomplished.		N
68.11	During the test described in 68.1, the primary switch is to be in the on position, both sides of the primary circuit of the appliance are to be connected together and to one terminal of the test equipment, and the second test equipment terminal is to be connected to the accessible dead metal.		N

69	**Grounding Continuity Test**	**No Grounding used.**	**N**
69.1	Continuity of grounding connection		N
69.2	Electrical indicating device		N

70	**Hair Dryer Power Input Test**		**N**
70.1	The power input to each hand-supported hair dryer shall be tested as described in 70.2, as a routine production-line test.		N
70.2	The power input is to be measured with the dryer at operating temperature under full-load conditions while connected to a circuit maintained		N
70.3	The results are acceptable if the power input to a dryer is within the inclusive range of 90 – 110 percent of the rating.		N
70.4	With regard to 70.3, the wattage of a dryer that has its electrical rating marked only in amperes and volts will be assumed to be the product of those two values.		N

RATINGS			
71	**Details**		**P**
71.1	An appliance shall be rated in volts and amperes or watts. It may be rated for alternating current only or direct current only.	125V	P
71.2	The current rating of an appliance shall include 15 amperes for a single receptacle provided as part of the appliance and intended as a general use outlet, 20 amperes for two or more receptacles (including a single duplex receptacle), or,		N

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Clause	Requirement + Test	Result - Remark	Verdict

	if the outlet is marked as noted in 72.1.6, that marked rating shall be included in the current rating of the appliance.		
71.3	The voltage rating of a cord-connected appliance intended for connection to a nominal 120-volt supply circuit shall not exceed 125 volts. The voltage rating of a dual-voltage appliance intended for connection to a nominal 120/240 volt supply circuit shall not exceed 125/250 volts.		P

MARKINGS			
72	**Details**		**P**
72.1	General		P
72.1.1	An appliance shall be legibly and permanently marked, where it will be plainly visible (after installation in the Case of a permanently connected appliance), with:		P
	a) The manufacturer’s name, trade name, trademark, or other descriptive marking by which the organization responsible for the product is identified;	See the making plate.	P
	a) The manufacturer’s name, trade name, trademark, or other descriptive marking by which the organization responsible for the product is identified;		P
	c) A distinctive (catalog) (model) number or the equivalent; and	See the making plate.	P
	d) The electrical rating.	See the making plate.	P
72.1.2	The repetition time cycle of a date code shall not be less than 10. The date code shall not requirereference to the manufacturer’s records to determine when the appliance was manufactured.		P
72.1.3	If a manufacturer produces or assembles appliances at more than one factory, each appliance shall have a distinctive marking, which may be in code, to identify it as the product of a particular factory.		N
72.1.4	A marking shall be:		P
	a) Paint-stenciled, die-stamped, molded, or indelibly stamped;		N
	b) In the form of pressure-sensitive labels; or		N
	c) In a form that has been determined to be the equivalent.		P
72.1.5	Block lettering shall be used for the marked words″ CAUTION,″″WARNING,″or″DANGER,″and for the marking wording that follows any of these words		N
72.1.6	An appliance provided with general-use receptacles intended for limited current loads shall have each such receptacle permanently marked″ amperes, maximum, watts, maximum″ or the equivalent, adjacent to the receptacle.		N
72.1.7	With regard to 7.9(i)(1), the convenience receptacle shall be permanently marked″ Amperes maximum, Watts maximum″or the equivalent adjacent to the convenience receptacle.		N
72.1.8	Each individual heating element or unit that is replaceable in the field shall be marked plainly in accordance with the requirement specified in 72.1.1.		N
72.1.9	Cleaning of the appliance or similar servicing by the user involves the exposure of any normally enclosed or protected live part to unintentional contact, the appliance shall be plainly marked to indicate that all such servicing is to be		N

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Clause	Requirement + Test	Result - Remark	Verdict
	done with the appliance disconnected from the supply circuit.		
72.1.10	An appliance having field-wiring terminals rated for supply connection using aluminum wire shall be marked with the following or the equivalent, as appropriate for the terminals.		N
72.2	Body- or table-supported hood- or bonnet-type hair dryers		N
72.2.1	A hand-, shoulder-, or table-supported hair dryer provided with a rigid hood or bonnet intended to be placed over or directly on the head while in use shall be marked in letters no less than 7/64 inch (2.8 mm) high with the word″WARNING″and the following or the equivalent:″To reduce the risk of possible electric shock do not use while bathing″.		N
72.3	Dual-voltage appliances		N
72.3.1	An input voltage selector shall be marked to indicate each individual voltage position.		N
72.4	Hand-supported hair-drying appliances		N
72.4.1	A hand-supported hair-drying appliance, such as a hair dryer, blower-styler, styler-dryer, heated air comb, curling iron-dryer combination, or similar appliance (with or without attachments) shall be permanently marked where readily visible in letters having a color contrasting with the color of the background:″DANGER – ELECTROCUTION POSSIBLE IF USED OR DROPPED IN TUB. UNPLUG AFTER USING″. The height of the letters in the word DANGER shall be no less than 1/8 inch (3.2 mm) and the height of the remaining letters shall be no less than 1/16 inch (1.6 mm).		N
72.4.2	An appliance of the type described in 72.4.1 shall be provided with a tag that is permanently attached to the power supply cord. The tag material and means of attachment to the power supply cord shall be judged under the requirements specified in the Test for Permanence of Cord Tag for Hand-Supported Hair-Drying Appliances, Section 58. The tag shall contain the following warning instructions.		N
72.4.3	The reverse side of the warning tag shall provide the pictorial warning illustrated in Figure 72.1, or the equivalent. The illustration shall consist of a black outline on a contrasting color background, with the slash mark in red. The height of the illustration shall be no less than 1 inch (25.4 mm) and the width shall be no less than 2 inches (50.8 mm). The heading″ UNPLUG IT″shall be in red letters at least 3/16 inch (4.8 mm) in height. The headings ″DO NOT REMOVE THIS TAG!″ and″WARN CHILDREN OF THE RISK OF DEATH BY ELECTRIC SHOCK!″shall be in black letters no less than 3/16 inch in height. All lettering shall be in block letters.		N
	(4.8 mm) in height. The headings ″DO NOT REMOVE THIS TAG!″ and″WARN CHILDREN OF THE RISK OF DEATH BY ELECTRIC SHOCK!″shall be in black letters no less than 3/16 inch in height. All lettering shall be in block letters.		
72.4.4	The warning tag shall be permanently affixed to the power		N

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Clause	Requirement + Test	Result - Remark	Verdict
	supply cord, no more than 6 inches (152.4 mm) from the attachment plug and shall be made of substantial material (cardboard, cloth, plastic, or the equivalent) to provide mechanical strength and to prevent easy removal.		
72.5	Permanently-installed wall-mounted hair dryers		N
72.5.1	A permanently-installed wall-mounted hair dryer shall be permanently marked either on the wall unit or on the hand unit where readily visible after installation in letters having a color contrasting with thecolor of the background:″DANGER – ELECTROCUTION POSSIBLE IF USED OR DROPPED IN TUB. TURN UNIT OFF AFTER USING″. The height of the letters in the word DANGER shall be no less than 1/8 inch (3.2 mm) and the height of the remaining letters shall be no less than 1/16 inch (1.6 mm).		N
72.5.2	A permanently-installed wall-mounted hair dryer shall be provided with a marking that is readily visible		N
72.5.3	The reverse side of the warning tag or the marking on the wall unit or on the hand unit shall provide the pictorial warning illustrated in Figure 72.2, or the equivalent.		N
72.5.4	The warning tag shall be permanently affixed to the cord of the hand unit, no more than 6 inches (152.4 mm) from the wall unit and shall be made of substantial material (cardboard, cloth, plastic, or the equivalent) to provide mechanical strength and to prevent easy removal.		N
72.5.5	A permanently installed wall-mounted hair dryer shall be marked on the wall unit or on the wall bracket where readily visible during installation with the word″ DANGER″ and the following or the equivalent:″To reduce the risk of death by electric shock, install only where hand unit cannot reach tub, sink, or shower″. The height of the letters in the word ″ DANGER″shall be no less than 1/8 inch (3.2 mm) and the height of the remaining letters shall be no less than 1/16 inch (1.6 mm). The letters shall be in a color which contrasts with the background. Block lettering shall be used for all words.		N
72.6	Curling irons		N
72.6.1	A hair curling iron or a heated comb shall be permanently marked, where readily visible, in letters no less than 7/64 inch (2.8 mm) in height with the word″WARNING″ and the following or the equivalent: ″To reduce the risk of possible electric shock do not immerse or use while bathing″or″Possible electric shock, do not immerse or use while bathing″.		N
72.6.2	A curling iron shall be provided with a marking on a tag that is non-permanently attached to the appliance or the power supply cord.		N
72.6.3	The reverse side of the tag specified in 72.6.2 shall provide the pictorial warning illustrated in Figure 72.4 including the cautionary statements shown.		N
72.7	Direct plug-in appliances		N
72.7.1	A direct plug-in appliance having a mounting tab shall be marked – on the appliance, a marking tag, or an instruction		N

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Clause	Requirement + Test	Result - Remark	Verdict
	sheet packed with the appliance – with the word ″CAUTION″and all of the following mounting instructions or the equivalent:		
	a)″To reduce the risk of electric shock – Disconnect power to the receptacle before installing or removing the appliance. When removing receptacle cover screw, cover may fall across plug pins or receptacle may become displaced.″		N
	b)″Use only with duplex receptacle having center screw.″		N
	c) ″Secure appliance in place by the screw provided with the appliance.″		N
72.8	Wax depilatory appliances		N
72.8.1	A wax depilatory appliance shall be legibly and permanently marked, in a location readily visible during intended use, with the following information. Each statement shall be separate and distinct. The height of the letters in the word″CAUTION″shall not be less than 1/8 inch (3.2 mm) and the height of the remaining letters shall not be less than 1/16 inch (1.6 mm). The lettering shall also comply with 72.1.4.		N
72.9	Appliances with GFCIs or similar protective devices		N
72.9.1	The surface of a plug that contains a GFCI or a similar protective device shall be marked with the word″WARNING″ and the following or the equivalent: ″To reduce the risk of electric shock, do not remove, modify, or immerse this plug. ″The height of the letters in the word″WARNING″shall not be less than 1/8 inch (3.2 mm) and the height of the remaining letters shall not be less than 1/16 inch (1.6 mm).		N
72.10	Convenience receptacle of a hand-supported hair dryer immersion protective device		N
72.10.1	A permanent and legible marking shall be provided near the convenience receptacle of an immersion protective device that is integral with the attachment plug of a hand-supported hair-drying appliance.		N
72.11	Hand-supported appliances with detachable power supply cords		N
72.11.1	A hand-supported appliance provided with a detachable power supply cord shall be permanently marked, where readily visible in letters no less than 7/64 inch (2.8 mm) in height, with the word “WARNING” and the following or equivalent: “To reduce the risk of electric shock use only with power supply cord provided with the appliance. ″		N
72.11.2	A hand-supported appliance provided with a detachable power supply cord shall be permanently marked, were readily visible in letters no less than 7/64 inch (2.8 mm) in height, with the wording “CAUTION – Shock Hazard. To provide continued protection against electric shock disconnect from the power supply when not in use.″		N

INSTRUCTION MANUALS			
73	**General**		**P**

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Clause	Requirement + Test	Result - Remark	Verdict
73.1	An appliance shall be provided with a user instruction manual that warns the operator of reasonably foreseeable uses or misuses so as to reduce the risk of fire, electric shock, or injury to persons. The instruction manual shall also consist of legible installation, operation, and, as applicable, user-maintenance instructions. The manual or other literature packaged with the product shall also indicate that the product is for household use. If included in the manual, the statement regarding intended use shall be in the operating instructions shown in Installation Instructions, Section 76.		P
73.2	The instructions pertaining to a risk of fire, electric shock, or injury to person		N
73.3	The instruction manual shall include instructions or illustrations to identify important parts of the appliance, such as the integral stand of a curling iron. An illustration may be used with a required written instruction to clarify its intent, but shall not be used in place of a required written instruction.		P
73.4	The following items shall be entirely in upper-case letters or shall be emphasized to distinguish them from the remainder of the text:		P
	a) The opening and closing statements of the		P
	instructions specified in 74.4 –″IMPORTANT SAFETY INSTRUCTIONS″or the equivalent,″READ ALL INSTRUCTIONS BEFORE USING,″ ″KEEP AWAY FROM WATER,″and″SAVE THESE INSTRUCTIONS;″		
	b) The headings″GROUNDING INSTRUCTIONS,″″ SERVICING OF DOUBLE-INSULATED APPLIANCES;″and		P
	c) The headings for the installation, operation, and user-maintenance instructions.		P
73.5	Unless otherwise indicated, the instructions shall be in the exact words specified or shall be in equally definitive terminology. Substitutes shall not be used for the words″WARNING″and″DANGER.″		P
73.6	Wording in parentheses in Sections 74 – 77 is explanatory, indicating options, alternatives, or cross-references. Wherever the words ″the (or this) appliance″ are used, the name of the specific appliance may be substituted in the final text.		P

74	**Instructions Pertaining to a Risk of Fire, Electric Shock, or Injury to Persons**		**P**
74.1	Instructions pertaining to a risk of fire, electric shock, or injury to persons shall warn the user of reasonably foreseeable risks and state the precautions that should be taken to reduce such risks. Such instructions shall be preceded by the heading ″ IMPORTANT SAFETY INSTRUCTIONS ″ or the equivalent.		P
74.2	The items in the list in 74.6 may be numbered and other instructions pertaining to a risk of fire, electric shock, or injury to persons that the manufacturer believes are needed may be included.		P

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Clause	Requirement + Test	Result - Remark	Verdict
74.3	The instructions shall be legible and contrast with the background. Upper-case letters in the instructions shall not be less than 5/64 inch (2.0 mm) in height and lower-case letters shall not be less than 1/16 (1.6 mm) in height. The heading ″ IMPORTANT SAFETY INSTRUCTIONS″or the equivalent and″SAVE THESE INSTRUCTIONS″shall be in letters not less than 3/16 inch (4.8 mm) in height .″READ ALL INSTRUCTIONS BEFORE USING,″″ GROUNDING INSTRUCTIONS,″″SERVICING OF DOUBLE-INSULATED APPLIANCES,″″WARNING, ″and″DANGER″shall be in letters not less than 5/64 inch in height, but less than 3/16 inch in height		P
74.4	The instructions pertaining to a risk of fire, electric shock, or injury to persons shall include those items in 74.5 (a) – (c) that are applicable to the appliance. The statement ″IMPORTANT SAFETY INSTRUCTIONS″or the equivalent,″READ ALL INSTRUCTIONS BEFORE USING,″and″KEEP AWAY FROM WATER″ shall precede the list in the order shown in 74.6, and the statement″SAVE THESE INSTRUCTIONS″orthe equivalent shall follow the list		P
74.5	The words″DANGER″ and″WARNING″ shall be entirely in upper-case letters or shall be emphasized to distinguish them from the rest of the text.		P
74.6	Two sets of instructions are shown for the sequence″ IMPORTANT SAFETY INSTRUCTIONS″ through the ″DANGER″ instructions.		N
74.7	As applicable, the following instructions shall be included in addition to the instructions in 74.6:		N
	a) For a facial steamer		N
	b) For a hair curler heater with water reservoir		N
	c) For a hair curler heater without water reservoir		N
	d) For a hair dryer that uses a liquid and is not hand supported		N
	e) For a hand-supported hair dryer, styling comb, or hair brush		N
	f) For a permanent wave appliance		N
	g) For a curling iron with water reservoir		N
	h) For a curling iron without water reservoir		N
	i) For a wig or brush dryer		N
	j) For a lotion heater		N
	k) For a dual-voltage appliance		N
	l) For a wax depilatory appliance		N
74.8	The instructions pertaining to a risk of fire, electric shock, or injury to persons shall include the instructions specified in 74.9 (a) – (f) as applicable.		N
74.9	The instruction manual shall include those instructions in (a) – (f) that are applicable to the appliance.		N

75	**Installation Instructions**	**N**

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Clause	Requirement + Test	Result - Remark	Verdict
75.1	Installation instructions shall contain all the information needed to install the appliance for use as intended, and shall be preceded by the heading ″INSTALLATION INSTRUCTIONS ″or the equivalent.		N
75.2	For an appliance provided with means for hanging, instructions regarding the correct installation of the hanging means intended for use shall be provided. See Use and Care Instructions, Section 91, for wall-hung hair dryers.		N
75.3	For a permanently installed wall-mounted hair dryer, the following instructions shall be provided: ″Install only where hand unit cannot reach tub, sink, or shower.″		N
75.4	If the mounting hardware specified in the Exception to 12.5 is not provided with the appliance, the instructions shall contain a description and an illustration of the hardware needed to mount the appliance.		N
76	**Operating Instructions**		**P**
76.1	Operating instructions shall contain all the information needed to operate the appliance as intended, and shall be preceded by the heading″OPERATING INSTRUCTIONS″or the equivalent. The operating instructions shall immediately follow the instructions specified in 74.9.		P
76.2	Operating instructions shall explain and describe the location, function, and operation of each user-operated control of the appliance, and warn against tampering with such devices. They shall also include the information specified in 76.3 – 76.7, as appropriate.		P
76.3	For an appliance using an automatically resetting thermal limiter that shuts off the entire appliance, instructions shall be provided to the user on what to expect in the event the thermal limiter operates and the appropriate action to take.		N
76.4	An appliance intended to be used with water, additives, conditioners, or other solutions with or without water, or an appliance that relies on the conductivity of water for intended operation (electrode-type-appliance), and for which the use of baking soda, salt, or other substances to improve the conductivity of the water is stipulated, shall be provided with specific instructions regarding the proper liquid or additive to use and the exact amount to be used (see 84.1).		N
76.5	For a dual-voltage appliance		N
76.6	For an appliance required to have a polarized plug, the following instructions shall be provided:″This appliance has a polarized plug (one blade is wider than the other). As a safety feature, this plug will fit in a polarized outlet only one way. If the plug does not fit fully in the outlet, reverse the plug. If it still does not fit, contact a qualified electrician. Do not attempt to defeat this safety feature.″		N
76.7	For an appliance of the type described in 72.4.1, instructions shall be provided for using the appliance (whenever it is used in bathrooms) on a circuit protected by a ground-fault circuit-interrupter.		N

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Clause	Requirement + Test	Result - Remark	Verdict
76.8	For a wax depilatory appliance		N
76.9	For an appliance provided with a GFCI or a similar protective device		N
76.10	With regard to 7.9(j)		N
76.11	With regard to the sequence of plugging and unplugging the hair dryer and the second appliance as specified in 76.10(d), the instructions shall specify that the appliance is to be plugged into the convenience receptacle of the immersion protective device of the hair dryer first, then the hair dryer is to be plugged into the wall receptacle outlet.		N
76.12	A curling iron shall be provided with an instruction as a permanent part of the instruction manual or as an insert card consisting of the pictorial warning as illustrated in Figure 72.4 including the cautionary statements shown.		N
76.13	Hand-supported appliances provided with a detachable power supply cord shall be provided with additional user instructions concerning insertion and removal of the detachable power supply cord, its storage, use and replacement		P

77	**User-Maintenance Instructions**		**P**
77.1	Instructions for user maintenance shall include explicit instructions for all cleaning and servicing – lubrication, adjustments, and the like – that are intended to be performed by the user, and shall be preceded by the heading″USER-MAINTENANCE INSTRUCTIONS″or the equivalent.		P
77.2	The instruction manual shall include a warning to the user that any other servicing should be performed by an authorized service representative or that the appliance has no user serviceable parts.		P
77.3	The instruction manual shall include specific instructions for the proper method of cord storage, total appliance storage, and the like when the appliance is not in use; and for cord care while in use (such as if the cord of a hand-supported appliance is twisted, to untwist the cord before use, or the like).		P

ELECTRODE-TYPE APPLIANCES			
78	**General**		**N**
78.1	A portable electrode-type appliance designed for household use on nominal 120-volt branch circuits shall comply with all applicable requirements in this standard and the particular requirements that follow. Accessories provided for electrode-type vaporizers or heating appliances are covered under requirements for those appliances.		N

79	**Construction**		**N**
79.1	An electrode-type facial steamer or attachment shall include a guard or barrier assembly (or the equivalent) to prevent emission of hot water droplets into the facial area of the appliance during intended		N

UL 859			
Clause	Requirement + Test	Result - Remark	Verdict

	use. The barrier assembly shall be constructed of materials acceptable for and mounted in the manner required for the enclosure of live parts.		
79.2	The interposed barrier assembly may consist of two or more layers arranged to provide a baffle effect for all openings.		N

80	**Operation Test**		**N**
80.1	An electrode-type facial steamer, when subjected to the operation test specified in 80.2		N
80.2	Three samples completely assembled as for intended use are to be subjected to the following test.		N
80.3	An electrode-type appliance other than specified in 80.1, when subjected to the operation test specified in 80.4		N
80.4	Three samples are to be subjected to the test indicated in 80.2.		N

81	**Leakage Current Test**		**N**
81.1	An electrode-type appliance shall be constructed so that, with the appliance assembled for use in its intended manner, there are no openings in the enclosure or guard permitting contact with the energized water by a copper alloy rod		N
81.2	The leakage current available from accessible surfaces (including those moistened by the liquid) under any condition of intended operation shall not be greater than 0.5 milliampere when the unit is operated with a hard water solution consisting of 1/2 gram of calcium sulfate, CaSO42H2O, per liter of distilled water.		N
81.3	During the determination specified in 81.1, 81.2, and 81.4, a barrier or enclosure whose acceptability for the particular application has not been determined is to be removed.		N
81.4	The leakage current, measured under the following conditions (a) and (b), shall not be greater than 0.5 milliampere		N
81.5	During the tests outlined in 81.2 – 81.4		N

82	**Disassembly and Reassembly**		**N**
82.1	If the instructions involve disassembly of any parts for cleaning, it shall be determined that the appliance is unlikely to be reassembled in a manner that will result in a risk of fire, electric shock, or injury to persons.		N

83	**Markings**		**N**
83.1	The appliance shall be marked with the following:		N
	a) A fill-level marking that can easily be compared with the actual water level during filling, or instructions for the use of any integral measuring container or other measuring means for filling. If the measuring container is not integral with the appliance, the amount shall be expressed in standard measurements in addition to the use of any measure provided.		N
	b) The word″CAUTION″and the following or the		N

UL 859			
Clause	Requirement + Test	Result - Remark	Verdict
	equivalent:″Shock hazard.		
	c) Instructions for filling, cleaning, and rinsing the appliance and any instructions on additives to be used.		N
	d) Instructions to keep the appliance out of reach of children.		N
	e) Instructions to locate the appliance where it will not be likely to be upset.		N
83.2	An electrode-type appliance having a water reservoir or boiling chamber with a capacity of 8 ounces (23.7 mL) or less shall be marked with the word ″ CAUTION″and the following or the equivalent:″To reduce the risk of excessive water temperatures that may cause burns if the unit is upset, follow manufacturer’s instructions on filling, cleaning, and rinsing.″ The height of the letters in the word ″CAUTION″shall not be less than 1/8 inch (3.2 mm), and the height of the remaining letters shall not be less than 7/64 inch (2.8 mm). For an electrode-type appliance, this marking shall be in addition to the marking specified in 83.1.		N
83.3	Cautionary markings and instructions shall be permanent and legible and shall be located on a part that cannot be removed without impairing the operation of the appliance.		N
83.4	Cautionary markings and instructions intended to instruct the operator shall be legible and clearly visible to the operator in the intended use of the appliance. Other such markings for servicing instructions should be legible and clearly visible when such servicing is being accomplished. Markings intended to reduce the risk of injury to persons shall be prefixed by the word″CAUTION″in letters no less than 3/32 inch (2.4 mm) in height.		N
83.5	A marking that is required to be permanent shall be:		N
	a) Molded, die-stamped, paint-stenciled, stamped, or etched on metal;		N
	b) Indelibly stamped lettering; or		N
	c) A pressure-sensitive label secured by adhesive that meets the requirements for the particular application (see 63.1 – 63.7). Ordinary usage, handling and storage of the appliance is to be considered in the determination of the permanence of the marking.		N

84	**Operating Instructions**		**N**
84.1	The manufacturer shall provide operating instructions with the appliance that shall include the following.		N

WALL-HUNG HAIR DRYERS			
85	**Scope**		**N**
85.1	These requirements cover cord-connected hair dryers rated 250 volts or less that consist of two non-detachable units– a hand unit and a wall unit with a length of flexible cord between the units.		N

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Clause	Requirement + Test	Result - Remark	Verdict

86	**General**		**N**
86.1	The appliance shall comply with the applicable requirements specified in Sections2–77andwith the requirements specified in Sections 87 – 91. The hand unit shall comply with the applicable requirements for a hand-supported hair dryer.		N

87	**Construction**		**N**
87.1	The appliance shall be provided with all the hardware necessary for hanging the wall unit in accordance with the installation instructions.		N
87.2	The wall unit shall engage the hanging means on the wall.		N
87.3	The appliance shall be constructed so that the hand unit cannot be energized while stored in the wall unit as intended.		N
87.4	The switch provided in accordance with 23.1.1 shall be in the wall unit, in the hand unit, or in both the wall unit and the hand unit.		N
87.5	A GFCI or other immersion protective device shall be integral with the attachment plug of the wall-hung hair-drying appliance power supply cord.		N

88	**Performance**		**N**
88.1	As part of the investigation, a wall-hung hair dryer shall be trial-installed to determine that the installation is feasible and that the instructions are detailed and correct.		N
88.2	The appliance shall comply with the Immersion Protection Trip Time Measurement Test, Section 40. The hand unit and wall unit shall be immersed simultaneously and the hand unit shall be off its holder. Power switches shall be tested in both the″on″position and the″off″position.		N

89	**Markings**		**N**
89.1	A wall unit or a hanging bracket shall be permanently marked where readily visible during installation in letters having a contrasting color to the background with the word ″DANGER″and the following or the equivalent:″To reduce risk of death by electric shock, Do not install where unit can fall into a tub or sink. ″The height of the letters in the word″DANGER″shall be no less than 1/8 inch (3.2 mm), and the height of the remaining letters shall not be less than 1/16 inch (1.6 mm). Block lettering shall be used for all words.		N
89.2	A warning tag that is in compliance with 72.4.2 – 72.4.4 shall be provided; however, use of following items are optional:		N
	a)″Always ’unplug it’ after use.″See 72.4.2(a).		N
	b) The heading″unplug it.″See Figure 72.1.		N

90	**Use and Care Instructions**		**N**
90.1	Warning instructions in compliance with Instructions		N

UL 859			
Clause	Requirement + Test	Result - Remark	Verdict

	Pertaining to a Risk of Fire, Electric Shock, or Injury to Persons, Section 74, shall be provided, with the following modifications		
	a) Item 1 under DANGER of Important Safety Instructions (see 74.6) shall be replaced with″Do not install unit where it can fall into a tub or sink. Always return hand unit to wall unit after using.″		N
	b) Item 1 under WARNING of Important Safety Instructions (see 74.6) need not be provided.		N
	c) The following statement or the equivalent shall be included after Item 9 under WARNING of Important Safety Instructions (see 74.6): ″Periodically inspect the wall unit for secure mounting.″		N

91	**Installation Instructions**		**N**
91.1	Installation instructions for hanging of the wall unit as intended shall be packaged with a wall-hung hair dryer. The instructions shall include a list of the items provided (such as screws, anchors, brackets,		N
	and similar hardware); the tools needed; and the step-by-step instructions for preparation of mounting surface, application of the hanging hardware, method of hanging the wall unit, and the like.		N
91.2	The installation instructions shall include the warning specified in 89.1, preceding the detailed instructions specified in 91.1.		N

SUPPLEMENT SA - HAIR DRYER ACCESSORIES			
SA1	Scope		N
SA2	Genera		N
SA3	Markings		N
SA4	Instructions		N

Appendix
Tables of Testing Data

Figure 7.1
Plug-face dimensions for determining acceptable convenience receptacle insertion clearance

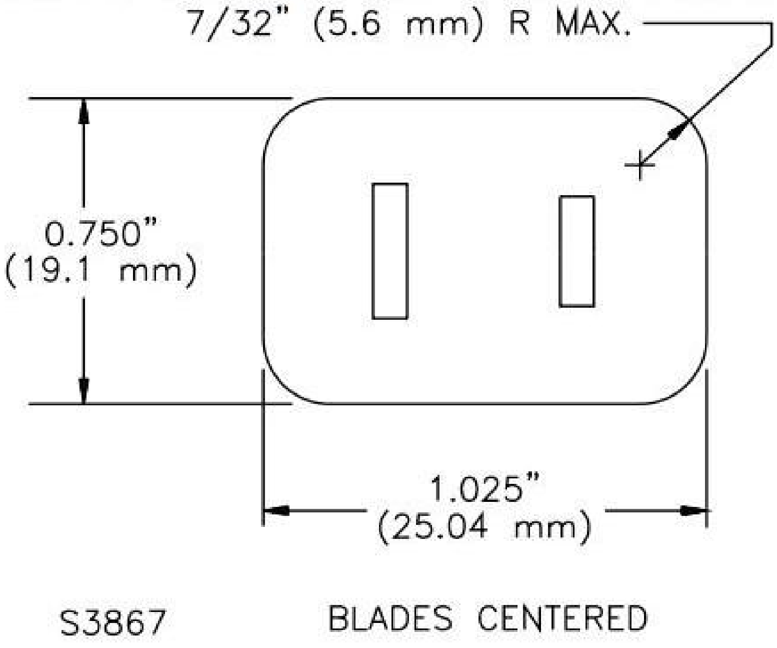
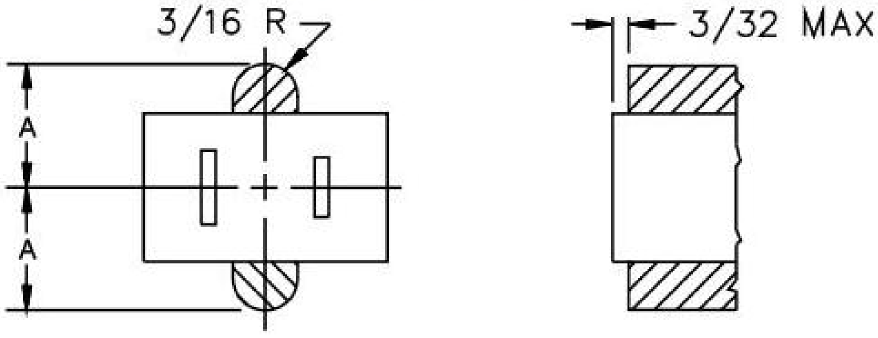
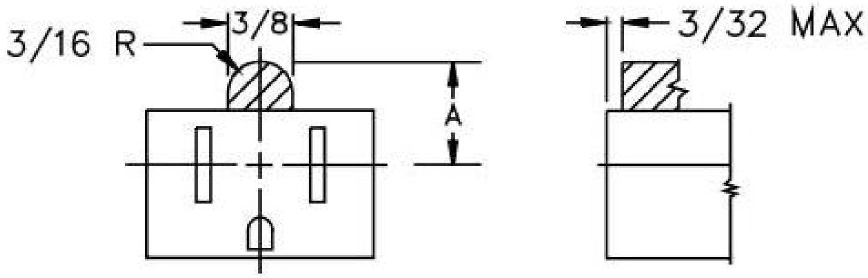


Figure 7.2
Face of a 15-ampere, 125-volt convenience receptacle showing the smallest acceptable obstruction (shown shaded) for the grounding pin on the mating plug



Obstruction for a 2-Pole, 2-Wire (Nongrounding) Receptacle



Obstruction for a 2-Pole, 3-Wire (Grounding) Receptacle

Inch	3/32	3/16
mm	2.4	4.8

Dimension A Inch	mm	Shore Durometer Hardness (Scale A)
0.625	15.9	less than 90
0.531	13.5	90 or more

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Table 13.1
Cord lengths for specific conditions

Type of appliance	Cord length			
	Minimum,		Maximum,	
	feet	(m)	feet	(m)
Appliance supported by hand or by table or counter top	5	(1.52)	9	(2.74)
Wall-hung appliances with attached appliance supported by hand	5[a] 2[b]	(1.52) (0.61)	9[a] 3[b]	(2.74) (0.91)
Appliance usually supported by head, shoulder, or back	6	(1.83)	12	(3.66)
Any appliance having a jacketed cord	See note[c]		Not specified	

[a] Between wall unit and hand unit.
[b] Between wall unit and receptacle.
[c] As specified elsewhere in this table or in 13.3.1.1.

Table 13.2
Required types of cord and applicable limitations on their use

Appliance on which the cord is to be used	Required cords where temperatures higher than 121°C (250°F) are attained on any surface the cord can touch	Required cords where 121°C (250°F) or lower temperatures are attained on any surface the cord can touch
Floor supported	HPN[a], HSJ, HSJO, HS, kHSO	SP-2[a], SPE-2[a], SPT-2[a], SV[a], SVE[a], SVO[a], SVOO[a], SVT[a], SVTO[a], SVTOO[a] SJ, SJE, SJO, SJOO, SJT, SJTO, SJTOO, S, SE, SO, SOO, ST, STO, STOO
Hand-supported hair dryers, heated air curling irons and brushes, and, except as noted in Note (b), counter-supported appliances	HPD, HPN, HSJ, HSJO	SP-2, SPE-2, SPT-2, SV, SVE, SVO, SVOO, SVT, SVTO, SVTOO, SJ, SJE, SJO, SJOO, SJT, SJTO, SJTOO, SP-1[b], SPE-1[b], SPT-1[b]
Facial saunas	–	SP-1, SPE-1, SPT-1
Combs	SP-2, SPE-2 or SPT-2, HPN	SP-2, SPE-2 or SPT-2, HPN

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Table 13.2 Continued

Appliance on which the cord is to be used	Required cords where temperatures higher than 121°C (250°F) are attained on any surface the cord can touch	Required cords where 121°C (250°F) or lower temperatures are attained on any surface the cord can touch
Curling irons and brushes, manicure and pedicure sets, hair crimping and hair straightening irons, and similar hand-supported appliances	TPT[c], SPT-1, XT[d], HPD	TPT[c], SP-1, SPE-1, SPT-1, XT[d], HPD

[a] Acceptable if the following conditions are met:

1) The unit is not provided with rollers, castors, or similar parts, and

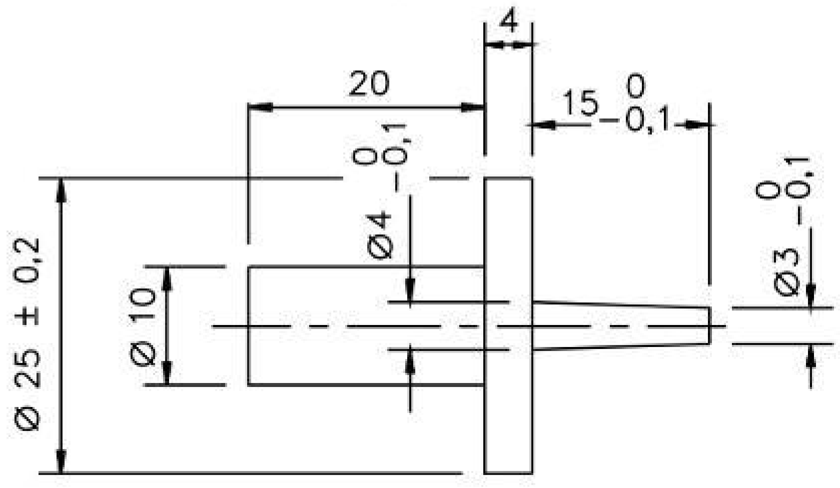
2) The point of cord entry to the unit is at least 3 feet (0.91 m) above the floor with the unit in any operating configuration.

[b] Acceptable on counter-supported appliances weighing 1/2 pound (0.23 kg) or less. The weight is to be determined without the power supply cord.

[c] Acceptable for use with appliances rated 50 watts or less and weighing 1/2 pound (0.23 kg) or less. The weight is to be determined without the power supply cord.

[d] Minimum 20 AWG (0.52 mm^2) parallel 2-conductor construction required.

Figure 13.3
Probe



Dimensions in millimetres

S2962D

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Tables of Testing Data

Table 25.1
Overvoltage categories

Appliance	Overvoltage category
Intended for fixed wiring connection	III
Portable and stationary cord-connected	II
Control located in low-voltage circuit	I
NOTE – Applicable to low-voltage circuits if a short circuit between the parts involved may result in operation of the controlled equipment that would increase the risk of fire or electric shock.	

Table 25.2
Material group

CTI PLC value of insulating materials	Material group
CTI ≥ 600 (PLC = 0)	I
400 ≤ CTI < 600 (PLC = 1)	II
175 ≤ CTI < 400 (PLC = 2 or 3)	IIIa
100 ≤ CTI < 175 (PLC = 4)	IIIb
NOTE – PLC stands for Performance Level Category, and CTI stands for Comparative Tracking Index as specified in the Standard for Polymeric Materials – Short Term Property Evaluations, UL 746A.	

Table 25.3
Pollution degrees

Appliance Control Microenvironment	Pollution degree
No pollution or only dry, nonconductive pollution. The pollution has no influence. Typically hermetically sealed or encapsulated control without contaminating influences, or printed wiring boards with a protective coating can achieve this degree.	1
Normally, only nonconductive pollution. However, a temporary conductivity caused by condensation may be expected. Typically indoor appliances for use in household or commercial clean environments achieve this degree.	2
Conductive pollution, or dry, nonconductive pollution that becomes conductive due to condensation that is expected. Typically controls located near and may be adversely affected by motors with graphite or graphite composite brushes, or outdoor use appliances achieve this degree.	3

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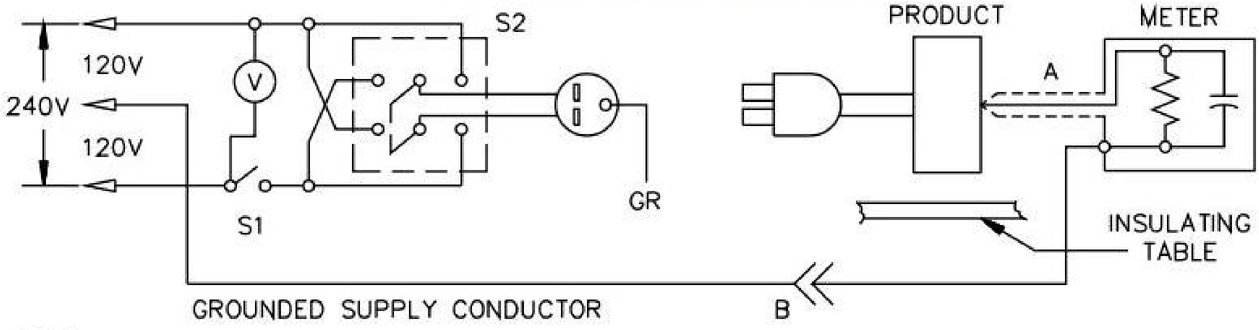
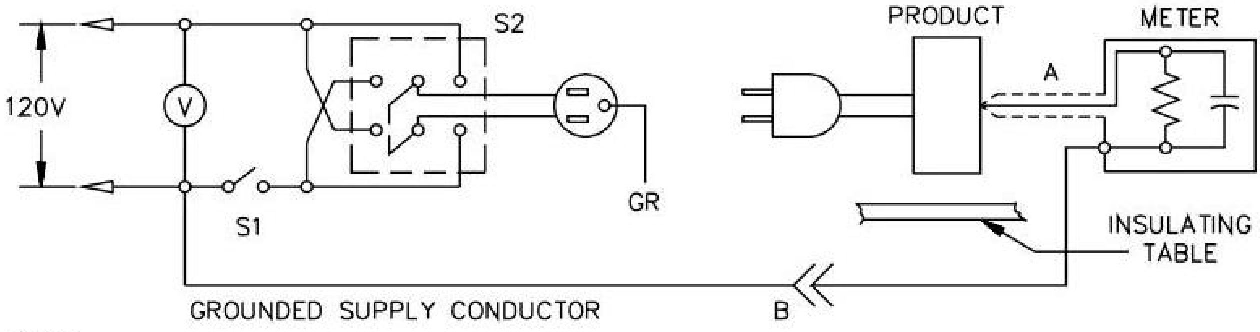
Table 28.1
Primary Class A insulating materials and minimum thicknesses

Material	Minimum thickness	
	Inch	(mm)
Vulcanized fiber	0.028	(0.71)
Polyethylene terephthalate film	0.007	(0.018)
Cambric	0.028	(0.71)
Treated cloth	0.028	(0.71)
Electrical grade paper	0.028	(0.71)
Mica	0.006	(0.15)
Aramid paper	0.010	(0.25)

Figure 38.1
Leakage current measurement circuits



LC200



LC100

A – Probe with shielded lead.
B – Separated and used as a clip when measuring currents from one part of the device to another.

Appendix

Tables of Testing Data

Table 45.1
Dielectric test potential

Appliance	Test potential, volts (rms)
An appliance not intended to be applied directly to a person, such as a wig dryer, hair curler heater (hair setter), hand-supported hair dryer, or similar appliance.	1,000
An appliance intended to be applied directly to a person without moisture present, such as a dry curling iron	1,000 + 2V[a]
An appliance (or attachment provided with it) intended to be applied in a wet or moist condition directly to a person, such as a steam curling iron, a mist-type hair curler, or similar appliance.	2,500[b]
An appliance such as a hair dryer-styler or hair untangler having comb or brush accessories, or both, that may be used for setting or styling of wet or damp hair	2,500[b]

[a] V is the maximum marked voltage but no less than 120 volts for a nominal 120-volt appliance or 240 volts for a nominal 240-volt appliance.
[b] A 2,500-volt potential is also to be applied to the handle, which is wrapped in metal foil, of a hand-supported appliance and to selected switch areas of any other appliance.

Table 60.1
Ball impact requirements for equipment

Portable				Stationary or fixed,	
Counter-supported,		Floor-supported,			
foot-pounds	(joules)	foot-pounds	(joules)	foot-pounds	(joules)
0.75	(1.02)	5.0	(6.8)	5.0	(6.8)

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2	TABLE: list of critical components and materials				P
object/part No.	manufacturer/ trademark	type/model	technical data	standard	mark(s) of conformity1)
ALCI	Zhongshan Rifeng Intelligent Electric Co. Ltd (E491497)	T15S	125V, 10A. Connected with power cord.	--	UL
	Zhongshan Rifeng Intelligent Electric Co., Ltd (200100444SHA-001)	T15S	125V, 10A. Connected with power cord.	--	ETL
	Foshan Shunde Xinhuida Electronic Co Ltd (E488714)	XHD15S02	125V, 10A. Connected with power cord.	--	UL
Power cord	Various	SPT-2	Rated 18AWG x 2C, 105ºC. Length: >1.52m and < 2.74m.	--	UL, ETL
Main switch	Shunde District of Foshan City Yimingda Electrical Company Limited (E508019)	KND-1	Rated: 125V, 16(4)A, T105, 1E4	--	UL
Internal wire	Various	1332	Rated: 300V~, 200ºC, 20AWG, 22AWG, 24AWG Used for connecting internal components.	--	UL, ETL
Heating Wire	Shunfa Electric Heating Material Co.,Ltd.	Ocr25AL5	R1: Φ0.42mm, R2: Φ0.25mm, 125VAC,1200W	--	Tested with appliance
Thermal cutoff	ZhongShan Qilin Electronics Co LTD (E526054)	QLF139	125V, 10A, 142°C. Secured to mica board by rivets.	--	UL
Thermal cut-out	ZHONGSHAN CHUANCHENG PRECISIONELE CTRONICS CO LTD (E465826)	CCS9	Rated: 125V, 20A, 10000 CYCLES, Tf130°C	--	UL
	FOSHAN SHUNDE ZHUO JIAN HARDWARD & ELECTRICAL MFG CO LTD (E248035)	ZJT-1B	Rated: 125V, 15A, 10K, Tf130°C	--	UL, ETL
Ionizer	Cixi Honge Electric Appliances (Intertek report 180700135SHA-001)	FF-370Rc	Input:100-130Vac, 60Hz, 1W; Output: -2.0 ~ -5.0KVdc.	--	ETL

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Ionizer	CIXI HONGE ELECTRIC APPLIANCES CO LTD (E325240)	FF-370Rc	Input:100-130Vac, 60Hz, 1W; Output: -2.5 ~ -4.5KVdc.	--	UL
Motor	FOSHAN SHUNDE YINGFUDA MICRO MOTOR CO., LTD.	YFD365V	DC30V, Class A	--	Tested with appliance
Main handle enclosure	CELANESE INTERNATIONAL CORP (E41938)	70G33HS1L(+)	PA66. Rated: HB, HWI=4, HAI=0, RTI (140, 125, 140), min. thickness 1.5 mm	--	UL
PCB	Various	Various	V-0, T130, Min. thickness: 1.5mm. Physically fitted into front enclosure.	--	UL

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38	TABLE: Leakage Current Test		P
	Supply Condition	I/P:125V~, 60Hz	--
Test voltage applied between:		Test Current (mA)	Allowed Current (mA)
Live parts L/N pin to metal enclosure		0.038/0.042	0.5
Live parts L/N pin to plastic enclosure		0.017/0.014	0.5
Remark:			

39	TABLE: Leakage Current Test Following Humidity Conditioning		P
	Supply Condition	I/P:125V~, 60Hz	--
Test voltage applied between:		Test Current (Ma)	Allowed Current (Ma)
Live parts L/N pin to metal enclosure		0.043/0.051	0.5
Live parts L/N pin to plastic enclosure		0.022/0.018	0.5
Remark:			

43	TABLE: Power Input Test			P
Input Condition	Rated power (W)	Input power (W)	Deviation (%)	Allowed Deviation (%)
125V / 60Hz	1200	1145.76	-4.52%	≤+5% or 20W And -10%
Remark:				

44	TABLE: Normal Temperature Test			P
	Supply Condition	AC 125V/60Hz		--
	T1(℃)	24.5		--
	T2(℃)	--		--
Temperature rise dT of part/at:		Test(℃)	Test(℃)	Required Tmax(℃)
Input wire		32.4	--	105
Internal wiring		44.6	--	175
Protection plug		30.5	--	45
PCB boards		48.1	--	105
Heating plastic support		60.2	--	125
Inside plastic enclosure		42.1	--	105
Outside plastic enclosure		38.5	--	60
Push button switch		32.6	--	60
Remark:				

45	TABLE: Dielectric Voltage-Withstand Test		P
Test voltage applied between:		Test voltage (V)	Breakdown
L and N to plastic enclosure		1250Vac	No
L and N to metal enclosure		1250Vac	No

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Remark：1 min.		

46	Table: Abnormal Operation Tests			P
ambient temperature (℃)..:		24.5		---
Component No.	Fault	Test voltage	Test time	Result
Motor	Stuck rotor	AC 125V/60Hz	10 min.	Unit shut down immediately; No High Temperature; No flammability gas; No hazard.
Supplementary information:				

Appendix A:
Equipment list

Code	Name	Model/Type	S/N	Calibrated date	Next Calibration Date	Manufacture
HTT-001	Digital Multimeter	34401A	MY47043456	2025.02.20	2026.02.19	agilent
HTT-004	Push/pull gSepe	NK-500	2Q10060932	2025.02.20	2026.02.19	--
HTT-005	Electronic weight	DSI-861	198692	2025.02.20	2026.02.19	shangdeli
HTT-006	Insulation resistance tester	CS2676CX	1107032-009	2025.02.20	2026.02.19	changshen
HTT-007	Earthing resistance tester	YD2668-4B	4B-2307	2025.02.20	2026.02.19	Yangzi
HTT-008	HI-pot/Insulation tester	CS2672C	1108006-002	2025.02.20	2026.02.19	changshen
HTT-010	AC Voltage Regulator	TDGC2J	--	2025.02.20	2026.02.19	SAKO
HTT-013	AC power source	HPA-3110	3513	2025.02.20	2026.02.19	Henqiang
HTT-014	Temperature/Humidity chamber	SDJ-80L	SDJ-80J	2025.02.20	2026.02.19	Shenzhen hongjian
HTT-015	Electric oven	HK45AS	F11011008	2025.02.20	2026.02.19	Guangzhou KENTON
HTT-017	AC digital power meter	PF9901	YG100731N11 070075	2025.02.20	2026.02.19	Yuanfang
HTT-019	DC electronic load	IT8512	002002506670 001002	2025.02.20	2026.02.19	ITECH
HTT-022	Leakage current tester	228	10-866030	2025.02.20	2026.02.19	simpson
HTT-023	Oscilloscope	TDS1012C-SC	C013300	2025.02.20	2026.02.19	tektronix
HTT-024	Tape measure	DK-2041	--	2025.02.20	2026.02.19	Proskit
HTT-025	Stop watch	TA-228	--	2025.02.20	2026.02.19	KTJ
HTT-026	Data acquisition/switch unit	34970A	MY44057668	2025.02.20	2026.02.19	Agilent
HTT-027	Temperature/humidity meter	VC230	--	2025.02.20	2026.02.19	ViCTOR
HTT-028	Torque drive	3RTD	435850B	2025.02.20	2026.02.19	TOHNICHI
HTT-030	Impact hammer	ZLT-CJ1	C011207	2025.02.20	2026.02.19	Guangzhou zhilitong
HTT-031	Inclined plane	ZLT-WD1	W011201	2025.02.20	2026.02.19	Guangzhou zhilitong
HTT-033	Test finger	ZLT-I02	I021203	2025.02.20	2026.02.19	Guangzhou zhilitong
HTT-034	Test pin	ZLT-I09	I091201	2025.02.20	2026.02.19	Guangzhou zhilitong
HTT-038	Test apparatus of the mains plug	ZLT-LJ2	LJ011202	2025.02.20	2026.02.19	Guangzhou zhilitong
HTT-039	Ball pressure apparatus	ZLT-QY1	Q011202	2025.02.20	2026.02.19	Guangzhou zhilitong
HTT-042	Caliper rule	CD-6 " CSX	500-196-20	2025.02.20	2026.02.19	MITUTOYO
HTT-044	Glow wire tester	ZRS-2	12121304	2025.02.20	2026.02.19	Guangzhou Xinna
HTT-045	Needle flame tester	ZY-2	12121311	2025.02.20	2026.02.19	Guangzhou Xinna

Appendix B
Figure documentation

PHOTO 1

[√] front
[ ] rear
[ ] right side
[ ] left side
[ ] top
[ ] bottom
[ ] internal

PHOTO 2

[√] front
[ ] rear
[ ] right side
[ ] left side
[ ] top
[ ] bottom
[ ] internal

PHOTO 3

[ ] front
[√] rear
[ ] right side
[ ] left side
[ ] top
[ ] bottom
[ ] internal

PHOTO 4

[ ] front
[ ] rear
[√] right side
[ ] left side
[ ] top
[ ] bottom
[ ] internal

PHOTO 5

[ ] front
[ ] rear
[ ] right side
[ ] left side
[ ] top
[ ] bottom
[ ] internal
[√] other

PHOTO 6

[ ] front
[ ] rear
[ ] right side
[ ] left side
[ ] top
[ ] bottom
[√] internal

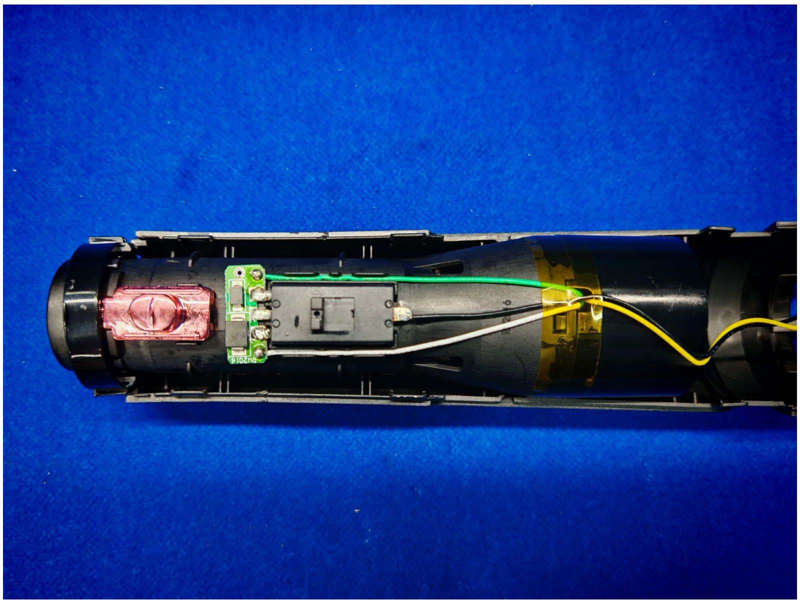
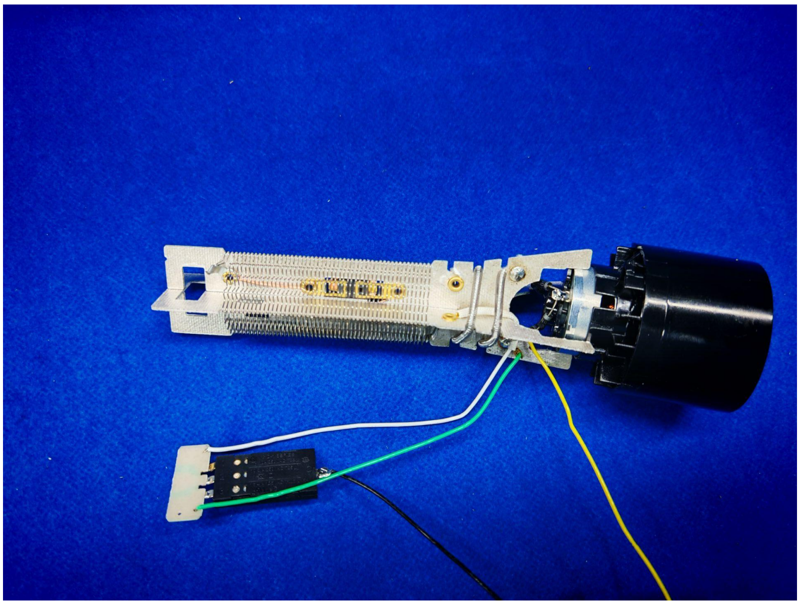


PHOTO 7

[ ] front
[ ] rear
[ ] right side
[ ] left side
[ ] top
[ ] bottom
[√] internal



*****End of the test report*****